

RiceMultiNet: Rice Leaf Disease Detection using CNN

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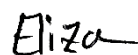
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Declaration

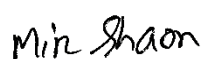
We do hereby declare that the research works presented in this thesis entitled, “**RiceMultiNet : Rice Leaf Disease Detection using CNN**” are the results of our works. We further declare that the thesis has been compiled and written by us, and no part of this thesis has been submitted elsewhere for the requirements of any degree, award, diploma, or any other purposes except for publications. The materials that are obtained from other sources are duly acknowledged in this thesis.



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I do hereby declare that the research works presented in this thesis entitled, “**RiceMultiNet : Rice Leaf Disease Detection using CNN**”, are the outcome of the original works carried out by the **Sajib Bapary , Md. Mir Shaon , Mirza Golam Habib, Eliza Akter , Md. Monir Hossain** under my supervision. I further declare that no part of this thesis has been submitted elsewhere for the requirements of any degree, award, diploma, or other purposes except for publications. I further certify that the dissertation meets the requirements and standards for the degree of Bachelor of Science in Computer Science and Engineering.



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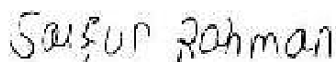
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Dedication

We would like to devote this research to our loving parents, who have been our constant source of strength through their prayers, support, encouragement, and love. They have sacrificed and guided and made us who we are today, and for that, we will always be grateful.

Acknowledgment

We would like to express our heartfelt gratitude to the almighty Allah who offered our family and us kind care throughout this journey. We are deeply thankful to the Bangladesh University of Business and Technology (BUBT) for providing us with such a wonderful environment to pursue our research. We would like to express our sincere gratitude to **Mr. Ali Azgar** , Assistant Professor, Department of CSE, BUBT. We have completed our research with his help. We found the research area, topic, and problem with his suggestions. He guided us with our study and supplied us with many research papers and academic resources in this area. He is patient and responsible. When we had questions and needed his help, he would always find time to meet and discuss with us no matter how busy he was. We also want to give thanks to **Md. Saiful Rahman** , Assistant Professor and Chairman, Department of CSE, and all our teachers for providing a solid background for our studies and research thereafter. They have been a great source of inspiration to us, and we thank them from the bottom of our hearts. We would also like to acknowledge our team members for supporting each other and be grateful to our university for providing this opportunity for us.

Abstract

Rice is a food staple crop, important for food security globally. Rice leaf diseases are an important constraint on crop yield, resulting in economic losses. The traditional diagnostic techniques are laborious and needed specialist. To tackle these problems, we are proposing a method using deep learning for the automatic identification of rice leaf diseases. Our study for Rice Leaf Disease Detection using Deep Learning was trained on a total of 17,480 balanced images that was divided into training (70%), validation (20%) and testing (10%). The dataset has variety of types of conditions on rice leaves that is Tungro, Sheath Blight, Rice Hispa, Neck Blast, Narrow Brown Spot, Leaf Scald, Leaf blast, Brown Spot and Bacterial leaf blight as well as healthy. We trained and tested eight deep learning models VGG19, ResNet50, DenseNet121, InceptionV3, Xception, MobileNetV2, EfficientNetB0, and EfficientNetB1 to detect different types of rice leaf diseases. The ResNet50 (97.59%), MobileNetV2 (99.37%), and EfficientNetB1 (99.86%) were the three models that worked best. We combined these three models to make a new ensemble model called RiceMultiNet because they all did very well on their own. The combined model did better than the single models, with an overall accuracy of 99.73%. The RiceMultiNet model that was suggested can find and sort ten different kinds of rice leaf diseases. This method makes it easier to find diseases, which helps farmers protect their crops and get more out of them.

Keywords: Rice Leaf Disease, Deep Learning, RiceMultiNet Model, Image Classification, Transfer Learning, Ensemble Model, Precision Agriculture.

List of Figures

F_No.	Figure Title	Page No.
1	External structure of rice plant	13
2	Motivation to improve Agricultural Practices through Deep Learning	16
3	Proposed structure for rice leaf disease classification	32
4	The Rice Leaf Dataset (Kaggle) Distribution of Classes	33
5	All Rice Leaf Disease Sample Image	34
6	Bar chart (Original vs Balanced class)	36
7	Pie chart or percentage distribution comparison	37-38
8	Class wise Image Distribution (Train, Val, Test)	39
9	Architecture of RiceMultiNet model	41
10	Comparison of Confusion matrix of MultiNet_Riceleaf architecture with others	48-56
11	Training and validation accuracy and loss curves of the RiceMultiNet model	57
12	Accuracy comparison of baseline models and the proposed RiceMultiNet model.	58
13	Rice Leaf Disease Detection Project Schedule	63

List of Tables

T_No.	Table Title	Page No.
1	Literature Review Summary table	27-29
2	Quantity of Original and Balancing Data	35
3	Data Augmentation Methods and Parameters	39-40
4	RiceMultiNet Model Architecture Summary	43
5	Summary of MultiNet/Ensemble-Based Rice Leaf Disease Detection Studies	59

List of Abbreviations and Acronyms

Abbreviation	Full Form
CNN	Convolutional Neural Network
IoT	Internet of Things
GAP	Global Average Pooling
ReLU	Rectified Linear Unit
SDG	Sustainable Development Goals
GPU	Graphics Processing Unit
IEEE	Institute of Electrical and Electronics Engineers
SHD	Sustainable Human Development
LSTM	Long Short-Term Memory
ML	Machine Learning
RGB	Red, Green, Blue (color channels)
LR	Learning Rate
VGG19	Visual Geometry Group Network (19 layers)
ResNet50	Residual Network (50 layers)
DenseNet121	Densely Connected Convolutional Network (121 layers)
InceptionV3	Inception Network Version 3
Xception	Extreme Inception Network
MobileNetV2	Mobile Network Version 2
EfficientNetB0	Efficient Neural Network (Baseline 0)
EfficientNetB1	Efficient Neural Network (Baseline 1)
VGG19	Visual Geometry Group Network (19 layers)
ResNet50	Residual Network (50 layers)
DenseNet121	Densely Connected Convolutional Network (121 layers)
InceptionV3	Inception Network Version 3

Table of Contents

Declaration	ii
Approval	iii
Dedication	iv
Acknowledgment	v
Abstract	vi
List of Figures	vii
List of Tables	viii
List of Abbreviations and Acronyms	ix
1. Introduction	1-20
1.1. Problem Statement	13-14
1.2. Problem Background	14
1.3. Research Objectives	15
1.4. Motivations	15-16
1.5. Significance of the Research	16
1.6. Research Contribution	16-17
1.7. Complex Engineering Analysis	17-18
1.7.1 Complex Engineering Problems	18-19
1.7.2 Complex Engineering Activities	20
1.8. Thesis Organization	20
2. Background Study	21-29
2.1. Literature Review	22-27
2.2. Problem Analysis	30
3. Research Methodology	31-44
3.1. Proposed Framework	32
3.2. Data Set Analysis	32-40
3.2.1. Data Preprocessing	40
3.3. Model Analysis	41-42
3.4. Integration of Multi-Scale Feature Representation	41-42
3.4.1. Pre-Trained Models	
3.4.2. Lightweight Models	
3.5. Pipeline for Model Tuning And fusion	42-44
3.5.1. Hyperparameter Settings	

4. Implementation and Result Analysis	44-59
4.1. Experimental setup	44
4.2. Model Implementation	44-45
4.3. Evaluation Metrics	45-47
4.4. Results	47-57
4.4.1. Training and parameter assessment	
4.5. Performance Analysis	57-58
4.6. Key Findings and Discussions	58-59
4.7. Comparison with State-of-the-Art Model	59
5 Standards, Constraints, and Milestones	60-63
5.1 Sustainability Standards	60
5.2 Impacts on Society	60-61
5.3 Ethics	61
5.4 Challenges	61-61
5.5 Constraints	61-63
5.6 Timeline and Gantt Chart	63
6. Conclusion	64-65
6.1 Limitations	
6.2 Future Works	

References

Appendix

Chapter 1

1. Introduction

Rice is a crucial staple crops in the world, especially in Asia, however it is prone to a range of diseases that impact the quantity and quality of the crop. Traditional methods for disease detection, which mainly consist of manual examination, are not only time-absorbing but also fail to provide accurate results, and this becomes especially troublesome in large scale rice farming. Deep learning model, especially Convolutional Neural Networks, can be found as a more accurate and faster solution for rice leaf disease automatic detection process.

Recent years have seen rapid advances towards using deep learning approaches to recount rice disease detection. As described in [1], proposed a hybrid deep-learning model for rice leaves classification, and gave an accuracy of over 92.68% accuracies on such dataset consisted of 6,086 images. Many other research workers like [2] employed optimization techniques to enhance the performance of their model, In [3], the authors combined multiple deep learning methods to achieve higher classification accuracy. In [4], Hybrid models were also MobileNet in disease detection accuracy of 94%. As demonstrated in [5] Using CNN and fusion techniques, [7] attained almost 100% accuracy in disease ID (bacterial blight, blast, tungo and brown spot) in rice.

In comparison with traditional rice diseases detection methods, CNN based machine learning models have proved to be means better. Such algorithms can rapidly analyze broad sets of images, detecting disease symptoms at a high level of accuracy. For example, researchers in [6] The researchers achieved an accuracy of over 95% using machine learning for classifying the rice diseases Several studies [13], [18], [19], [24] and [30] High test accuracies of above 95% have also been achieved using deep learning techniques.

Furthermore, rice disease monitoring is also benefited by the innovation of new technologies, such as drones and IoT. According to previous studies [30] and [24] Examples include who reported classification accuracies of 97.6% and 99.43% for larger spatial scales of rice fields using CNN and SVM. With the help of these technologies, the detection of diseases will be faster and more efficient, closing the crop management gap.

In several rice-growing countries, like Bangladesh, disease outbreaks have been known to reduce harvests by as much as 30%, compromising food security [10]. The diseases such as leaf blast, bacterial blight, and brown spot cannot be detected in the early stages and environmental conditions like temperature, humidity, and rainfall have a great influence in the spread of such diseases [14]. These early signs are often missed by traditional methods and thus lead to a delay in intervention. Deep learning is a better, faster, and more accurate solution to it, as it will allow for early detection, thus preventing mass crop loss[1].

Deep learning is being utilized for rice disease detection, overcoming the boundary of traditional methods those are slow and subjective. Using convolutional neural networks (CNNs) and hybrid models to classify rice leaf images improve accuracy, giving farmers timely response to diseases (**Figure 1** shows the external structure of a rice plant). As mentioned in [12] and [13], Allowing for the integration of technologies (drones, IoT), machine learning models can be part of a bigger picture that brings greater benefits with more real-time data available for faster interventions.

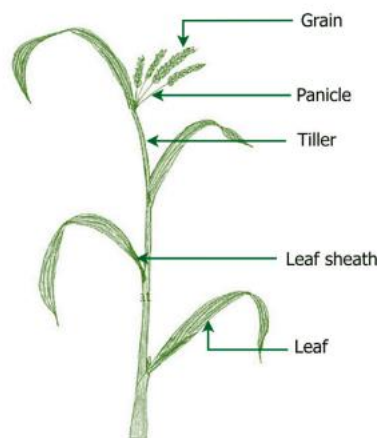


Figure 1: External structure of rice plant.

In short, I can say deep learning approaches, especially CNNs and hybrid models, can be a very effective tool for detecting leaf diseases in a rice plant. Such techniques are more robust and fast as compared to traditional approaches, which can assist in saving the crops and enhance rice agriculture mentioned in [5]. With advancement of technology, integration of deep learning with real-time data retrieval and emerging technologies such as drones will strengthen detection of diseases in rice for improving rice production and food security [11].

1.1. Problem Statement

Disease on rice leaves is closely related to the destruction of the rice crop, so it is a significant problem for rice farmers, as it has an impact on food security and farmers' income. Existing approaches for monitoring these diseases, such as experts physically examining their crops, are gradual, costly and impractical for larger farms. With these methods, there is a significant amount of time and expert knowledge involved which makes it hard for farmers to diagnose diseases at an early stage which is vital to prevent crop loss.

Machine learning and deep learning are the modern technologies that can aid in detection of these diseases, but there still exist challenges. Out of these, many models suffer from accuracy issues while identifying various categories of rice leaf disease. Also, current models may not support a high volume of data or work in real-time scenarios According to previous studies [3] and [7].

It is required a much-prompt and accurate approach for identification of rice leaf diseases at real-time and will require much less human involvement using a highly sophisticated deep learning model. It should work good under different conditions and be able to effectively classify the different rice leaf diseases reported in [5] and [8].

This research work is trying to overcome these challenges through a deep learning model that increases the accuracy of the diseases detection and can be applied easily in actual agricultural settings. The aim is to enable farmers to detect diseases earlier and cultivate their crops more effectively.

1.2. Problem Background

Numerous illnesses and environmental issues are posing a growing threat to rice production, lowering both yield and quality. In order to overcome this obstacle and guarantee sustainable rice production and food security, early disease detection and efficient control are crucial.

Tungro, Sheath Blight, Rice Hispa, Neck Blast, Narrow Brown Spot, Leaf Scald, Leaf Blast, Brown Spot and Bacterial Leaf Blight are multiple diseases that severely think rice farming through lowering crop production and its quantity One of the significant challenges faced by rice farmers is dealing with disease outbreaks that can devastate their harvest. Grids of thousands of fields can not be managed as farm owners spent several hours working in farms for inspecting their farms by themselves as diseases detection methods require a lot of expert knowledge and take a long time. Deep-learning of models in detecting disease is a potential

solution, permitting early diagnosis with high precision. In fact, despite the improvements, the best model for detecting multiple rice leaf diseases is still an open problem. Existing solutions have taken approaches that revolve around using models such as VGG, ResNet, DenseNet, model and several fusion architectures, with mixed results with respect to accuracy, precision and general applicability.

1.3. Research Objectives

These traditional methods usually give average accuracy for Rice leaf disease detection. Our ultimate objective is to give a better precise approach to detecting diseases in Rice leaves. Other goals include:

- ❖ **Environment / Leaf Condition Agnostic Disease Detection** : Create a model that can detect diseases consistently, under different environmental conditions or leaf conditions. together with VGG19, ResNet50, DenseNet121, InceptionV3, Xception, MobileNetV2, EfficientNetB0, EfficientNetB1 Hybrid Model and their accuracy in rice leaf disease classification.
- ❖ **Disease Classifier**: Classify different types of diseases based on their unique visual symptoms
- ❖ **Trend Analysis for Agronomists**: Population trends in disease progression and spread to help farmers and agricultural planners.

1.4. Motivations

Rice is also a staple food for millions and a common livelihood for many farmers as well. Diseases that affect rice leaves, on the other hand, can damage rice and decrease yield, resulting in economic losses. These diseases need to be controlled to ensure the health of the crops through their early and accurate detection.

Of course, traditional approaches of finding out the diseases are by examining the plants manually, as we often say, to discover and find there hand by hand, which is too much time-consuming and also human error prone. Fortunately new technologies, and particularly deep learning, can offer a helping hand. Such techniques will contribute to rapid and accurate analysis of leaf images and can facilitate improved disease management and more sustainable farming. The aim of this study is to apply deep learning (DL) technique to enhance the detection of rice leaf diseases from plant-leaf images to strengthen agriculture and food security we show in **Figure 2**.

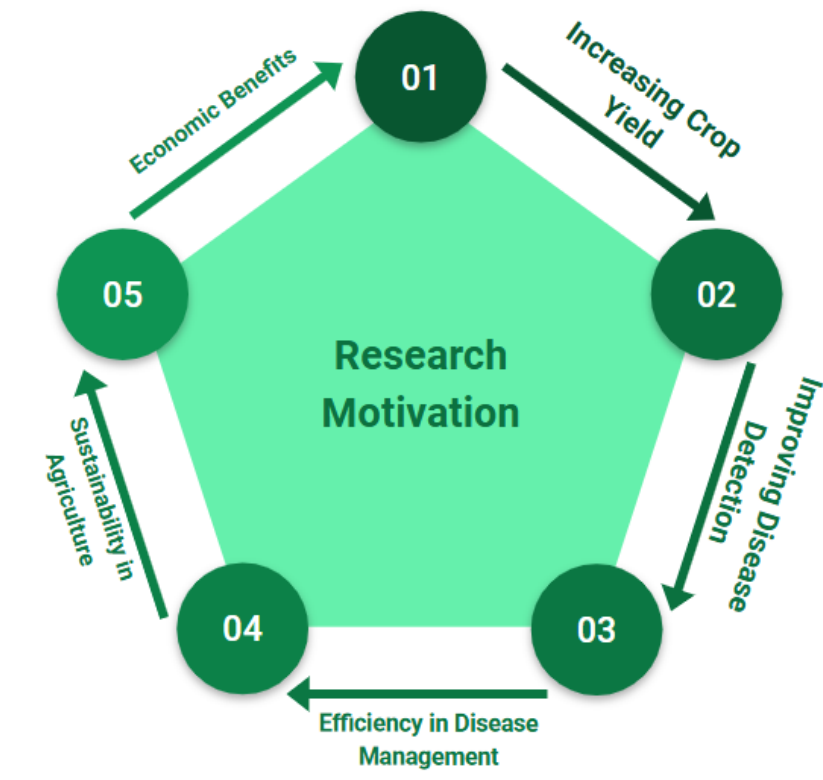


Figure 2: Motivation to improve Agricultural Practices through Deep Learning.

1.5. Significance of the Research

Introduction there is an increasing demand in detection of rice leaf diseases in efficient timely manner and scalable solution. The research has the following objectives by employing deep learning methods:

- ❖ **Improve Productivity:** By detecting diseases at an early stage rather than a later stage, farmers avoid damaging large parts of the crop and are able to keep healthier plants yielding more.
- ❖ **Promote Sustainable Agriculture:** The model help sustain agriculture as it reduces the use of chemical treatments thus resulting in lesser impact on the environment.
- ❖ **Enhance Food Security:** Improving rice, a staple food crop for billions of people, has a direct impact on global food security.

Farmers can now act quickly and improve their farm management for minimal losses, which is especially important in developing countries where rice is considered a staple food.

1.6. Research Contribution

- ❖ **Model Assessment:** We tested and compared various deep learning models for the classification of rice leaf disease and applied VGG19, ResNet50, DenseNet121, InceptionV3, Xception, MobileNetV2, EfficientNetB0, EfficientNetB1 models.
- ❖ **Novel Approach with Pre-Trained Models:** The study emphasizes the utility of pre-trained models for image classification in agricultural settings, that have been lesser explored.
- ❖ **Dataset creation and analysis:** A diverse dataset was utilized for the study, covering a broad range of rice leaf diseases, with data augmentation techniques employed to enhance model robustness.

This research proposes a model that would provide farmers with an early detection of the disease, eliminating the need for manual inspection and improving the decision-making process.

1.7. Complex Engineering Analysis

This part of your thesis focuses on the technical challenges of using deep learning to detect diseases in rice leaves. The main goal is to understand the issues that come with implementing this model in real-world farming conditions.

1. Data Preprocessing and Dataset Selection

Deep learning models need a lot of data to work well. But not all data is perfect. Some datasets may have images that are blurry, have different sizes, or don't represent all types of rice leaves and diseases. This can cause problems, making the model perform poorly. To solve this:

- ◆ A **diverse dataset** will be used, which includes both healthy and diseased rice leaves.
- ◆ **Data augmentation** techniques will be applied, like rotating or flipping images, to make sure the model sees many different types of images, helping it learn better.
- ◆ Studies like [1] used large datasets with over 6,000 images, so handling large datasets is important to ensure good performance.

2. Model Selection and Hybrid Architecture

Choosing the right model is key to success. Some research shows that combining multiple models can improve accuracy. For example:

- ◆ **Inception-ResNet-V2** achieved 92.68% accuracy reported by [1].
- ◆ A **custom deep neural network (DNN)** with a special algorithm reached 96.96% According to [2].
- ◆ **Hybrid models**, which mix different types of models (like CNNs), can improve prediction accuracy even more. The challenge here is to combine the best models to get the best results without using too much computing power.

3. Accuracy vs. Computational Efficiency

Deep learning models, especially **CNNs**, need a lot of computing power to train and run. But some models, like **VGG-19** and **EfficientNet**, can offer a good balance between **accuracy** (how correct the model is) and **computational cost** (how much power is needed to run it).

In your research, combining three strong models, like **EfficientNetB**, **MobileNetV2** and **ResNet50**, will create a multinet model. This model aims to deliver high accuracy while being efficient enough to run on devices like smartphones or drones.

4. Model Evaluation and Deployment

After building the model, it needs to be tested to ensure it works well in real-world conditions. The model will be evaluated using several performance metrics, such as:

- ◆ **Precision** (how many of the predicted diseased leaves are actually diseased).
- ◆ **Recall** (how many of the actual diseased leaves were correctly identified).
- ◆ **F1 score** (a balance between precision and recall). Studies like [7] stress the importance of testing the model on new, unseen data to make sure it doesn't just memorize the training data but generalizes well. Additionally, the model must work under different conditions, like changes in **lighting** or **humidity**, which can affect image quality.

1.7.1. Complex Engineering Problems

❖ P1: Depth of Knowledge Required

- This project demands expertise in deep learning and image recognition (K8), along with practical knowledge in plant disease pathology (K3).

- Advanced skills in image processing and neural network design (K7) are needed for optimal model development.
- Data engineering skills for handling large datasets (K4) and implementation skills for field deployment (K5 & K6) are essential.

❖ **P2: Conflicting Data Requirements**

High-quality data is needed, but real-world datasets often include incompatibilities, such as poor lighting or background noise. This can impact the accuracy of the model and present challenges in data analysis.

❖ **P3: Depth of Analysis**

The project requires a thorough integration of various factors, such as disease severity, leaf angle, and environmental conditions, to create a robust and coherent model that can operate across different scenarios.

❖ **P4: Familiarity of Issues**

Disease symptoms may not have clear or direct correlations, making it difficult to create a straightforward classification model. This requires careful analysis and feature extraction to capture the nuances in visual data.

❖ **P5: Use of Advanced Methods**

The project will leverage deep learning models, including convolutional neural networks (CNNs) and potentially generative approaches, to enhance the robustness of disease detection.

❖ **P6: Stakeholder Involvement**

This research will involve collaboration with agricultural scientists, farmers, and agribusiness experts who need accurate and user-friendly disease detection tools. Their feedback will help refine the model and its application.

❖ **P7: Interdependence**

- Plant Pathology and Detection
- Big Data Computation

1.7.2. Complex Engineering Activities

❖ A1: Range of Resources

1. High-quality images.
2. Agronomic Data.
3. Environmental Factors.
4. Comprehensive Dataset.

❖ A2: Level of Interaction

Dealing with complex, real-world data is challenging and requires advanced techniques to handle issues like changes in lighting or partial obstructions in images.

❖ A3: Innovation

The project needs creative methods to help models learn from messy and varied data, using strategies like data augmentation and transfer learning.

❖ A4: Societal and Environmental Benefits

Better disease detection can lower the need for excessive pesticide use, supporting more sustainable farming. Early detection also helps farmers increase their yields and improve economic stability.

❖ A5: Practical Experience

This project applies machine learning and deep learning to real-world agricultural problems, blending theory with hands-on solutions for real impact.

1.8. Thesis Organization

This thesis is divided into six chapters:

Chapter 1: Introduces the research problem, objectives, and importance of the study.

Chapter 2: Reviews previous studies and explains the research gap.

Chapter 3: Describes the proposed method, dataset, and model used.

Chapter 4: Shows implementation, results, and performance analysis.

Chapter 5: Discusses standards, challenges, and project timeline.

Chapter 6: Concludes the work with findings and future directions.

Chapter 2

2. Background Study

Based on the image, there are few studies that have made notable progress in improving rice leaf disease detection based on deep learning techniques. Islam et al. The authors [1] proposed a hybrid deep learning framework for paddy leaf diseases identification on the dataset of 6086 images and reported very high accuracy using Inception-ResNet-V2. As stated in [2] proposed the DNN-CSA model which employed deep neural networks and the Crow Search Algorithm, attaining a remarkable accuracy of 96.96%. According to [3] The Efficient Deep Learning based Fusion Model (EDLFM-RPD) created by showed 96.17% accuracy for rice plant disease detection on the dataset of 40 rice images.

Or more recently with Deep learning techniques, advanced deep learning models have shown quite promising results in this field, Registers such as VGG19 and ResNet and CNN have shown accuracy rates of 87% to 98% in some cases As described in [4–10]. A recent study by [8] integrated CNN with IoT to establish a multi-model fusion framework, resulting in 95.31% accuracy. Meanwhile, [7] used the YOLOv5 model with Kaggle rice leaf images to identify four common pathologies with a precision of 1.00.

Innovative approaches including transfer learning, GANs and hybrid models have also been used in some studies in order to improve detection accuracy, as seen in the work of [11] and [18], where models achieved precision recall and F1 scores collectively approaching 100% for rice leaf disease classification. These developments showcase the ability of deep learning to transform the field of agriculture through effective realtime disease diagnosis mechanisms.

2.1. Literature Review

Islam et al. [1] Their research created a hybrid architecture for paddy leaf disease detection that is based on deep learning. 984 samples of both healthy and diseased paddy leaves were included in the dataset. After analyzing four models VGG19, Inceptionv2, ResnetV2, ResNet101, and Xception it found that Inceptionv2 and ResNetV2 had higher accuracy, 92.68%.

Nalini et al. [2] This work presents a DNN, CSA model that uses the Crow Search Algorithm to identify paddy leaf diseases accurately and efficiently. It achieves a classification accuracy of 96.96%, which is a significant improvement over conventional SVM models and provides useful assistance for farmers.

Almasoud et al. [3] The article provides the Efficient Deep Learning based Fusion Model (EDLFM-RPD), which was created using a dataset of 40 images of rice plants in order to detect and classify rice plant diseases with an effective classification accuracy of up to 96.17%.

Mahrin Tasfe et al. [4] The present work explores deep learning approaches for early detection and classification of paddy diseases, focusing on major models, datasets, and augmentation techniques. It reviews strategies to enhance model performance and identifies gaps and challenges, providing insights for precision agriculture and guiding future research in sustainable rice disease management.

Vasanthan et al. [5] Recent developments in deep learning and machine learning methods for early rice disease detection are covered in the paper. It highlights deep feature-based SVM methods showing superior F1-score and accuracy, AlexNet neural networks achieving 0.99 accuracy, and CNN models with high level fusion techniques achieving perfect accuracy of 1.0 for common diseases.

Ibrahim et al. [6] The study examines the application of machine learning to rice plant disease detection. They worked together using a dataset of 1,000 high-resolution photos of rice leaves, including samples that were both healthy and infected. The accuracy of the researchers' model is over 95%. It outperforms more conventional techniques like J48, Naïve Bayes, and SVMs with an accuracy rate of over 95%.

Jhatial et al. [7] The YOLOv5 deep learning model, which we use in this study, was trained on a dataset of 400 images of rice leaves that were gathered from Kaggle. It is capable of accurately classifying four common rice leaf diseases—Bacterial Blight, Blast, Brown Spot, and Tungro

with precision scores of 1.00, recall of 0.94, and mAP of 0.62, indicating its potential for use in real-time agricultural practices in the future to increase farm output.

Kumar et al. [8] In this study, a multi-model fusion framework called RICE Fusion is used. which uses CNN combined with IoT-based agricultural information to accurately diagnose rice leaf diseases. The framework was tested using 3200 data sets and they Achieved 95.31% test accuracy. Which is CNN 82.03% and MLP 91.25% so good results that it outperforms unimodal models.

Kulkarni et al. [9] This project highlights the effectiveness of deep learning in rice leaf disease diagnosis. A dataset of 1649 images of four classes of rice leaves is used here. The CNN model showed exceptional performance with 95% test confidence. This includes dividing the different image collections into training-based training (70%) and testing 30% subsets, and processing techniques are used to improve reliability. With support vector machines such as (SVM or KNN) evaluated with various classifiers, the decision trees achieved 95% maximum inefficiency on test data.

Kondaveeti et al. [10] Rice leaf disease detection has been extensively researched using image processing, machine learning, and deep learning techniques. Earlier methods used handcrafted feature extraction, while recent studies have adopted deep learning models like YOLO, U-Net, DenseNet, ResNet, and VGGNet for automated detection. Hybrid approaches, including CNN+RNN, CNN+Autoencoder, and ML+DL, have improved accuracy in handling complex disease patterns.

Kathiresan et al. [11] The authors provide a transfer learning model soberingly accurate to detect rice leaf diseases while leveraging generative adversarial network (GAN) for sample diseased image balancing. It has been found out to attain placing 98.79% average cross-validation accuracy on gan-augmented datasets and 98.38% on three other datasets. The study further reveals the problems with the quality of images aplenty and brings up some suggestions regarding extending the range of steps up to disease types and low-computation models creation. At the same time, it demonstrates the usefulness of the datasets created using GAN and evaluates GAN with trained models built on other datasets.

Yang et al. [12] The present work proposes the Dense Higher Level Composition Detection Transformer (DHLC, DETR), which applies the Detection Transformer (DETR) model for the detection of flax spots, rice blast, and sheath blight in rice leaf disease. It incorporates a Dense Higher Level Composition Feature Pyramid Network (DHLC, FPN) and increases the mean

Average Precision to 17.3%, for small targets to 9.5%, and the average accuracy to 97.44% for the IDADP dataset.

Latif et al. [13] In order to achieve their primary goal of early recognition improving crop output, we present in this article an improved Deep Convolutional Neural Network (DCNN) finding model based on VGG19 with the use of drone-based and Internet of Things (IoT) technology for segmentation. The system successfully detected and classified rice leaf diseases with an accuracy of 96.08% using a non-normalized augmented dataset.

Krishnamoorthy et al. [14] Rice is considered an integral source of food security around the world. Leaf blasts, bacterial blight, and brown spots are some of the diseases that have been shown to affect its productivity and thus require a real time and accurate approach for rice leaf diseases diagnosis. The research applies transfer learning to utilize existing convolution neural network models VGG-16, ResNet50, and InceptionV3 for diagnosing these depicted, achieving a classification rate of 87%, 93% and 95% respectively.

Khasim et al. [15] Convolutional neural networks and other machine learning techniques, such as leaf smut, bacterial leaf blight, and brown spot disease detection, achieved more than 97% accuracy with a dataset of 240 images using decision tree algorithms and CNN models between 0.54 and 0.93. This study highlights the need for an automated intervention for the identification and management of rice plant diseases in Bangladesh.

Zein et al. [16] In this study, the Nu Support Vector Machine (NuSVM) algorithm is used to classify rice leaf diseases using a segmented rice leaf dataset. The algorithm achieves a respectable accuracy (52.12% to 53.81%) on five cross validation folds, with the goal of improving precision, recall, and F1 score for better diagnosis provisioning.

Anandhi et al. [17] The ESHODL MFRPDC method proposed in this paper which is based on deep learning, computer vision and optimization techniques reaches high accuracy 98.77% and ideal values also for precision 97.21% and sensitivity 97.65% in identifying rice plant diseases. The method achieves better performance and efficiency with minimal CPU computation time (0.12 sec) than other deep learning models.

Amjad et al. [18] It proposes a hybrid model for rice leaf disease detection that combines CNN, LSTM, and SA modules. An important contribution to agricultural disease identification and growth monitoring, the model achieved 100% precision, recall, and F1 score for all classes, with 97-98% efficiency in test data and 100% efficiency in training samples.

Ahad et al. [19] A customized VGG16 model based on DL and transfer learning techniques is proposed in this paper, which automates the highly accurate detection and classification of rice leaf diseases. The suggested approach outperformed current desktop station models in terms of prediction accuracy (99.94%) under nine disease classifications, as demonstrated by its validation on a dataset of 5932 images of rice leaves. It is also a workable approach to early disease detection for agriculture.

Mac et al. [20] The work suggests a new modified version of YOLOv8 for rice leaf disease detection that makes use of the EIoU and α -IoU loss functions. The model achieved 89.9% accuracy after being trained on a dataset of 3175 photos representing three disease classes (blast leaf, leaf folder, and brown spot). The modified YOLOv8 model provides an effective solution for real-time IoT-based agricultural disease detection by outperforming the state-of-the-art object detection techniques, according to a comparison with YOLOv7 and YOLOv5.

Raigonda et al. [21] In this work, a hybrid CNN-VGG model was proposed using transfer learning for rice leaf diseases (brown spot, leaf blast and leaf blight) classification which obtained an accuracy of 95.60% with 1800 images compared to the traditional CNN methods offering better performance.

Subbarayudu et al. [22] A CNN based deep learning Hybride model to separate 13,866 rice leaf disease images into categories of brown spot, leaf blast, healthy etc gave an accuracy rate as high as 98.52%, showing the potential in real-time agricultural field scenarios.

Arafath et al. [23] Several CNN models were used in this study on eight different plant species, using a total dataset of 30,945 photos taken from leaves and A specially created dataset of rice leaves 350 photos per class was used for the experiments. It included six cases: bacterial leaf blight, brown spot, healthy, leaf blast, leaf scald, and narrow brown spot. a real-time web/mobile application for effective disease identification and management, and up to 98% accuracy in identifying rice diseases (by DenseNet121).

Kunal et al. [24] This paper proposed using a dataset of 5,932 thermal augmented rice leaf images, accuracy in detecting the hybrid Darknet53 model with SVM model reached up to 99.43% in disease detection and showed great promise for real-time agricultural applications.

Martins et al. [25] In this study, a lightweight CNN model named RiceLeafClassifier-v10 was constructed. Trained on 2,807 images with five different rice leaf conditions at an impressive

94% of accuracy after quantization which is capable of running real-time smoothly on edge devices like Raspberry Pi.

Mehnaz et al. [26] In another study, the research compared CNN, ViT and SVM using Dhan-Shomadhan dataset for rice leaf disease detection in Bangladesh; with the SV ResNet50 method achieved an accuracy rate of 91.5% which shows that it work better than a transformer and tradisional models.

Baiju et al. [27] For the purpose of detecting diseases in corn, rice, and wheat crops, a thin deep learning model called Slender-CNN was introduced. This will work for many crops, but crop-specific models are different, making them impractical for farmers with low levels of digital literacy. It uses parallel convolutional layers of various sizes to accurately detect disease lesions at various scales. In addition to achieving an overall accuracy of 88.54%, the model outperformed other well-known CNN models, including VGG19, EfficientNetB6, ResNet101, DenseNet201, AlexNet, YOLOv5, and MobileNetV3. The rice accuracy rate is 87.11%. The Slender-CNN can be used practically in low-resource farm settings because it is small and uses fewer resources.

Kumar et al. [28] CNN-based deep learning has been used in this paper to predict rice diseases like bacterial blight, brown spot, and blast. 10,407 photos from 10 classes made up the dataset we used. The highest accuracy was 97.50% for DenseNet121, 96.32% for Xception, 96.25% for MobileNet V3 Large, and 96.32% for EfficientNet B4. Class Weight: This project employed the class weight method, which directly prioritizes undercounted diseases, such as bacterial panicle blight, during model training. It balances disease possession and enables more accurate detection, both of which are impossible to achieve with threshold adjusting.

Khatri et al. [29] highlights a deep learning-based technique for identifying and categorizing different kinds of rice leaf diseases. With a dataset of 13,960 annotated images from 9 classes (such as Hispa, Brown Spot, Leaf Blast, and Bacterial Leaf Blight), this study evaluated several CNN models as well as pretrained versions such as VGG19, EfficientNetB0, ResNet50, and DenseNet121. The best performing model was DenseNet121, which had 94% accuracy, 0.93 precision, 0.94 recall, and an F1-score of 0.93. Two public datasets were combined to create our dataset, and flipping and rotation techniques were used to enhance the images. Before the model was trained, the images were resized and normalized as follows: It provides a useful tool for automated diagnosis of rice pathogens without the need for manual inspection, which would be advantageous for sustainable agriculture.

Murugan et al. [30] Rice leaf diseases, such as blast, brown spot, tungro, and bacterial blight, were identified using deep learning, more especially CNN and nine transfer learning models. At 97.6%, DenseNet121 had the highest accuracy in classifying diseases. The 7,432 images in the dataset 5,932 of which were diseased and 1,500 of which were healthy were taken from the rice field and used to train the AI model. The results made a substantial contribution to the classification of diseases based on the accuracy and effectiveness of this AI. A synopsis of the previously discussed works is provided in brief.

Author(s)	Study Focus	Disease Class	Dataset	Model(s)	Accuracy
Islam et al.[1]	Hybrid deep learning architecture for paddy leaf disease detection	5 Types of Rice leaf	984 paddy leaf images	VGG19, Inceptionv2, ResNetV2, ResNet101, Xception	92.68% (Inception-ResNet-V2)
Nalini et al.[2]	DNN, CSA model with Crow Search Algorithm for efficient paddy disease identification	3 Types of Rice leaf	120 image	DNN-CSA	96.96%
Almasoud et al. [3]	Efficient Deep Learning based Fusion Model (EDLFM - RPD) for rice plant disease detection	3 Types of Rice leaf	40 rice plant images	EDLFM-RPD	96.17%
Tasfe et al. [4]	The review focuses on the strategies implemented to improve model performance and identifies any existing gaps or challenges.	15 Types of Rice leaf Disease	Not specified	Not specified	Not specified
Vasantha et al. [5]	ML and DL techniques for automated rice disease detection	6 Types of Rice leaf Disease	Not specified	CNN with high-level fusion, AlexNet, SVM	1.0 (CNN); 0.99 (AlexNet)
Ibrahim et al. [6]	ML techniques for detecting rice plant diseases	3 Types of Rice leaf Disease	1,000 rice leaf images	Not specified	95%
Jhatial et al.[7]	YOLOv5 model for detecting major rice leaf diseases	3 Types of Rice leaf Disease	400 rice leaf images	YOLOv5	Precision-1.00, Recall 0.94, mAP 0.62
Kumar et al. [8]	RICE Fusion framework integrating CNN and IoT for rice leaf disease diagnosis	4 Types of Rice leaf Disease	3200 rice leaf images	CNN, MLP	95.31%
Kulkarni et al. [9]	Deep learning for rice leaf disease diagnosis	4 Types of Rice leaf Disease	1649 rice leaf images	CNN, MLP,	95% (CNN)
Kondaveeti et al. [10]	Research has been conducted on rice leaf disease detection using image processing,	8 Types of Rice leaf Disease	Not specified	YOLO, U-Net, DenseNet, ResNet, VGGNet and	Not specified

	machine learning, and deep learning techniques.			Hybrid (ML+DL)	
Kathiresan et al. [11]	Transfer learning with GANs to balance rice leaf disease sample sizes	4 Types of Rice leaf Disease	GAN-augmented dataset 12,496 image	MobileNet, ResNet50, ResNet101, Inception V3, and Xception	98.79% (GAN dataset), 98.38% on three other datasets
Yang et al. [12]	Dense Higher Level Composition Detection Transformer (DHLC, DETR) for rice leaf disease identification	3 Types of Rice leaf Disease	IDADP dataset 600 image	DHLC-DETR	97.44%
Latif et al. [13]	Modified VGG19 with IoT and drone tech for early rice disease detection	6 Types of Rice leaf Disease	Non-normalized augmented dataset (2,167 image)	Modified VGG19	96.08%
Krishnamoorthy et al. [14]	Transfer learning for identifying common rice diseases	4 Types of Rice leaf Disease	1,000 image	VGG-16, ResNet50, InceptionV3	87%,93%,95 %
Khasim et al. [15]	Automated rice disease detection system for Bangladesh	4 Types of Rice leaf Disease	240 rice leaf images	J48, Decision Tree	97% (Decision Tree)
Zein et al. [16]	Classifying rice leaf diseases with Nu-SVM	3 Types of Rice leaf Disease	Not specified	Nu-Support Vector Machine (Nu-SVM)	52.12% - 53.81%
Anandhi et al. [17]	Deep learning method for rice leaf disease detection	5 Types of Rice leaf Disease	UCI database	ESHODL-MFRPDC (Deep learning, Computer Vision, Optimization)	98.77%
Amjad et al. [18]	Hybrid model (CNN, LSTM, SA) for rice leaf disease detection	4 Types of Rice leaf Disease	5932 rice leaf disease images	CNN, LSTM, SA Modules	97-98% test data, 100% training
Ahad et al. [19]	VGG16-based model with transfer learning for rice leaf disease detection	4 types of Rice leaf diseases	5932 rice leaf images	VGG16 (custom), Xception, ResNet50, DenseNet121, Inception ResnetV2, and Inception V3	99.94% (VGG16)
Mac et al. [20]	Modified YOLOv8 with EIoU and α -IoU loss for rice leaf disease detection	3 Types of Rice leaf Disease	3175 images	Modified YOLOv8	89.9%
Raigonda et al. [21]	Rice leaf disease classification	3 Types of Rice leaf Disease	1,800 images	Hybrid CNN-VGG (transfer learning)	95.60%
Subbarayudu et al. [22]	Rice leaf disease classification	9 Types of Rice leaf Disease	13,866 images	MobileNetV3-GKSO-SA-CatBoost	98.52%

Arafath et al. [23]	Multi-plant leaf disease classification including rice	7 Types of Rice leaf Disease	350 Image rice leaf dataset	custom CNN, VGG16, VGG19, InceptionV3, MobileNet, DenseNet121, Xception, and two hybrid models	98% (DenseNet121 for rice)
Kunal et al. [24]	Rice disease detection with thermal image augmentation	3 Types of Rice leaf Disease	5,932 Image	Hybrid Darknet53 + SVM	99.43%
Martins et al. [25]	Lightweight rice leaf disease classifier for edge devices	5 Types of Rice leaf Disease	2,807 images	RiceLeafClassifier-v1.0 (lightweight CNN)	94% (post quantization)
Mehnaz et al. [26]	Comparison of CNN, Vision Transformer (ViT), and SVM for rice disease	3 Types of Rice leaf Disease	5,200 image	ResNet50, ViT, SVM	91.5% (ResNet50 best)
Baiju et al. [27]	Multi-crop disease detection (corn, rice, wheat) with lightweight model	4 Types of Rice leaf Disease	Rice: 32 image Corn: 32 image Wheat:36 image	VGG19, EfficientNetb6, ResNeXt, DenseNet201, AlexNet, YOLOv5 and MobileNetV3	87.11% for rice(Proposed Model)
Kumar et al. [28]	Rice disease detection with class imbalance handling	10 Types of Rice leaf Disease	10,407 images across 10 classes	DenseNet121, Xception, EfficientNet B4, MobileNet V3	97.50% (DenseNet121 highest)
Khatri et al. [29]	Rice leaf disease detection and classification	9 Types of Rice leaf Disease	13,960 images across 9 classes	CNN, VGG19, EfficientNetB0, ResNet50, DenseNet121	94% (DenseNet121 best)
Murugan et al. [30]	Rice leaf disease classification with transfer learning	5 Types of Rice leaf Disease	5,932 images	VGG19, VGG16, DenseNet121, MobileNetV2, DenseNet169, DenseNet201, InceptionV3, ResNet152V2, and NASNetMobile	97.6% (DenseNet121 highest)

Table 1: Literature Review Summary Table.

2.2. Problem Analysis

One of the key challenges in Rice farming is the diseases caused by pests and pathogens, which can depress crop yields and threaten food security. As we have seen, early identification of these diseases is important, however doing that manually is time-consuming and challenging especially in expansive rice fields. This is the reason researchers are working to detect rice leaf diseases automatically using deep learning (AI).

There are multiple studies which proves this fact to be true with Deep learning application. For example, [1] achieved 92.68% accuracy for leaf disease detection in rice plants using hybrid model. Similarly, [2] 96.96% (despite employing varied methodology) But there are still some hurdles:

- ❖ **Inconsistent accuracy:** Different models produce different results. Some generalize well to one set of data but perform poorly when evaluated on other data [12, 16]. This means we have to find a better model.
- ❖ **Small Datasets:** Lots of models are trained on small datasets. For instance, Almasoud et al. This may lead to overfitting as [3] used only 40 images, leading the model to perform well on the training data (which it has already seen) but poorly on new data.
- ❖ **Use in Real Time:** Some models perform excellently in lab conditions, and ensuring their application in real-life farms with varying weather and light conditions is still a challenge [7]. We need to ensure the models are functioning appropriately in real-time farming context.
- ❖ **HIGH COMPLEXITY:** Deep learning models such as GANs [11] or complex networks [18] require immense computation and power. This makes them inaccessible in under-resourced rural regions.
- ❖ **IoT Integration:** Some models, when combined with IoT (smart devices in farming), can yield better results. Still this integration is done in a early stage and we must make sure that it works in practice, i.e., with data coming from physical sensors and other devices [8].

In this study, we seek to address these issues by developing a more comprehensive yet simple model that functions in real-time and is applicable in varying conditions of farming. The importance is how do you make it practical and affordable for the farmers.

Chapter 3

3. Research Methodology

The work flow of detecting rice leaf diseases, displayed in **Figure 3**

Our database of rice leaf images was significant and diverse. In this stage we took the time to clean and remove low quality or corrupted images from the dataset. We further applied multiple preprocessing steps to enhance image quality and diversity. These steps were balancing the dataset, reducing noise and performing image augmentation (resizing, rotating and flipping) to provide more thorough training of the models.

The dataset was pre-processed followed by splitting the data to training, validation and test. We trained and evaluated the following state of the art deep learning models/models-architectures Custom CNN, VGG19 [17], ResNet50 [18], PyTorch Lightning, DenseNet121 InceptionV3, Xception, MobileNetV2, EfficientNetB0[28][29], and finally a Hybrid Model (Approach used to combine the goodness of multiple architectures).

This was trained separately for each model to find the category of rice leaf disease images. The accuracy, precision, recall and F1-score In [19], were used to evaluate their performance in detecting particular rice leaf diseases accurately.

Our final and most effective approach is of Hybrid Model which has shown finer performance than individual models. Eventually, the goal is to deploy this model in agriculture fields and even pair it with IoT devices for real-time monitoring, disease forecasting at earlier levels of infection and better crop management aiding farmers in maximizing yield whilst minimising loss As reported by [20].

3.1. Proposed Framework

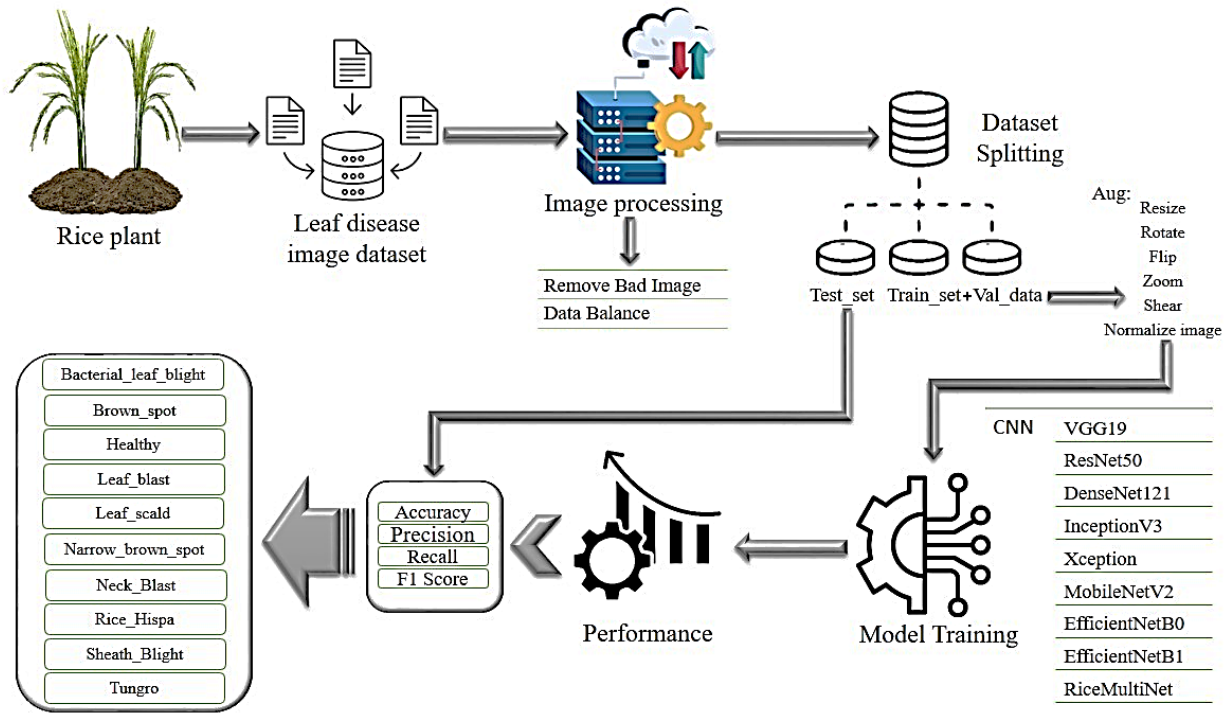


Figure 3: Proposed structure for rice leaf disease classification.

3.2. Data Set Analysis

Dataset Disease Types & Descriptions

It has developed this for 10 rice leaf diseases namely Sheath Blight, Rice Hispa, Neck Blast, Narrow Brown Spot, Leaf Scald, Leaf Blast, Healthy, Brown Spot, Bacterial Leaf Blight and Tungro were used. In **Figure 5**, sample images of rice leaf dataset Provided with background. The 13,530 images in the dataset; Used the data from Kaggle with the next attributes: In **Figure 4**, we describe the number of data in the dataset. **Figure 7**, Displays the percentage distribution of images in the original dataset based on the class-wise distribution. In the field of rice leaf disease detection, various types have been extensively studied using deep learning models, as discussed in several referenced studies. Bacterial Leaf Blight, a prevalent disease, has been effectively identified by CNN models like VGG19 and DenseNet121, noted in references [5], [11], [17] and [19] for their adept handling of complex image classifications. Similarly, Brown Spot has been analyzed with robust detection capabilities provided by models such as ResNet50 and YOLOv8, referenced in [5], [11], [18] and [20].

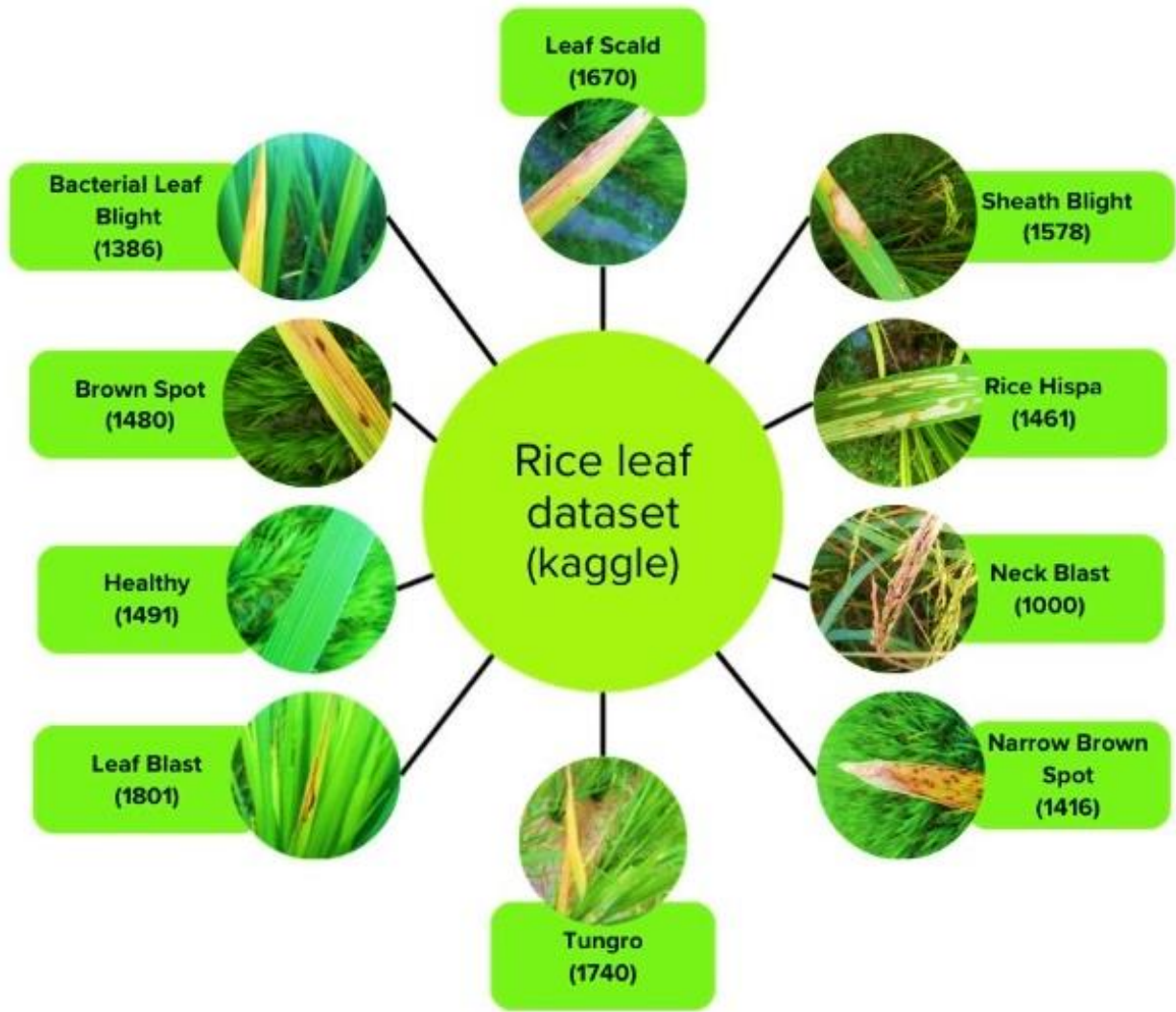


Figure 4: The Rice Leaf Dataset (Kaggle) Distribution of Classes

The importance of identifying Healthy plants, as outlined in references [11] and [19], serves as a benchmark for assessing the effectiveness of disease detection models. Leaf Blast and Leaf Scald are detected using advanced CNN architectures and augmentation techniques, improving accuracy in classification tasks, as shown in references [11], [17] and [19] for Leaf Blast and [11], [17] for Leaf Scald. Narrow Brown Spot, Neck Blast, and Rice Hispa utilize hybrid models combining CNN and LSTM frameworks, along with IoT technology for real-time monitoring, highlighted in references [11], [18], [19] for Narrow Brown Spot, [11], [18] for Neck Blast, and [17], [18], [20] for Rice Hispa. Sheath Blight and Tungro are also critical diseases, with Sheath Blight being detected by DenseNet121 among other models, providing high accuracy as cited in [5], [11], [17], [20]. Tungro detection benefits from ensemble learning models that enhance detection rates, as mentioned in [5], [11] and [18]. These references

collectively emphasize the advancements in agricultural technology and disease management, showcasing the significant potential of deep learning in improving crop health and productivity.



Figure 5: All Rice Leaf Disease Sample Image

3.2.1. Data Preprocessing

Data preprocessing plays a pivotal role in ensuring the efficacy of our models. In this stage we took the time to clean and filter out any bad images from the dataset. To achieve balanced representation and minimize model bias, we adjust the original data to create a "Balanced Original Data" All classes were balanced with maximum possible sample size. Our augmentation techniques, which include resizing (224, 224) images to uniform dimensions, rotating, flipping, zooming, shearing, and normalizing pixel values, are designed to enhance model robustness by introducing variability that mirrors real world scenarios. Finally, we split the processed dataset into training, validation, and test sets to streamline the training process, fine tune model parameters, and assess performance on new, unseen data.

Data balancing

Before training the model, we inspected the dataset and observed that some of the classes had a large number of images while others had very few of them. This is the reason why the model might lean towards larger classes and to address this data balancing was performed. To begin, all images were resized to 224×224 . Then, the smaller classes were oversampled with transformations (such as Flip, Left, Right) As long as the maximum class data set is equal. In this manner, we obtained a balanced data set of all classes. The model was given equal opportunity to learn the features of all classes, thereby reducing bias. As we can see in **Figure 6** demonstrates that the original dataset had significant class size differences, but after balancing, the distribution became even, in **Figure 7** we examine the percentage distribution of classes before and after balancing. The largest class had 12.9% and the smallest 7.1%, but when balancing, all classes contributed the same (10%/1,748) of the data, with 1,748 images representing the total number of classes, and we can see in **Table 2** Quantity of Original and Balancing Data. The dataset is divided into training, validation, and testing sets with a 70-20-10% ratio after balancing. Which we can see in **Figure 8** how the spelling is completed.

Rice Leaf Disease Class	Original_ Data	Balancing_Data
Sheath Blight	1629	1748
Rice Hispa	1299	1748
Neck Blast	1000	1748
Narrow Brown Spot	954	1748
Leaf Scald	1332	1748
Leaf Blast	1748	1748
Healthy	1085	1748
Brown Spot	1546	1748
Bacterial Leaf Blight	1197	1748
Tungro	1740	1748
Total	13,530	17,480

Table 2: Quantity of Original and Balancing Data.

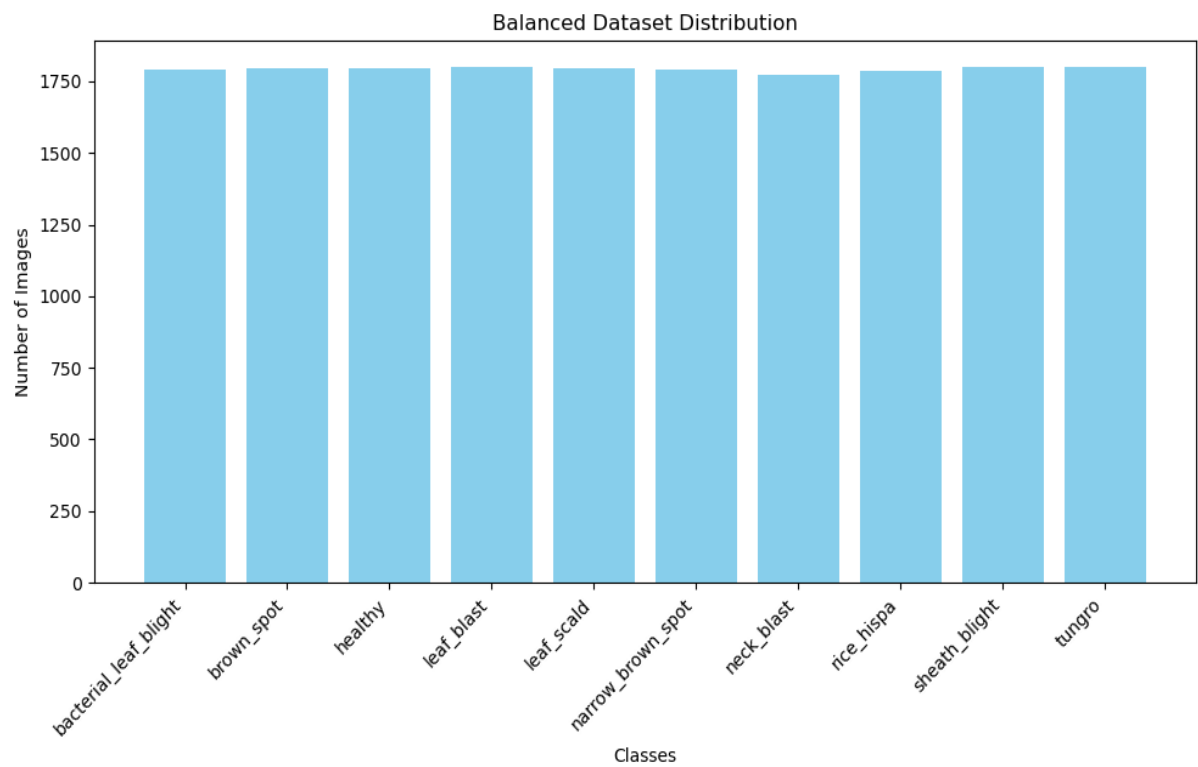
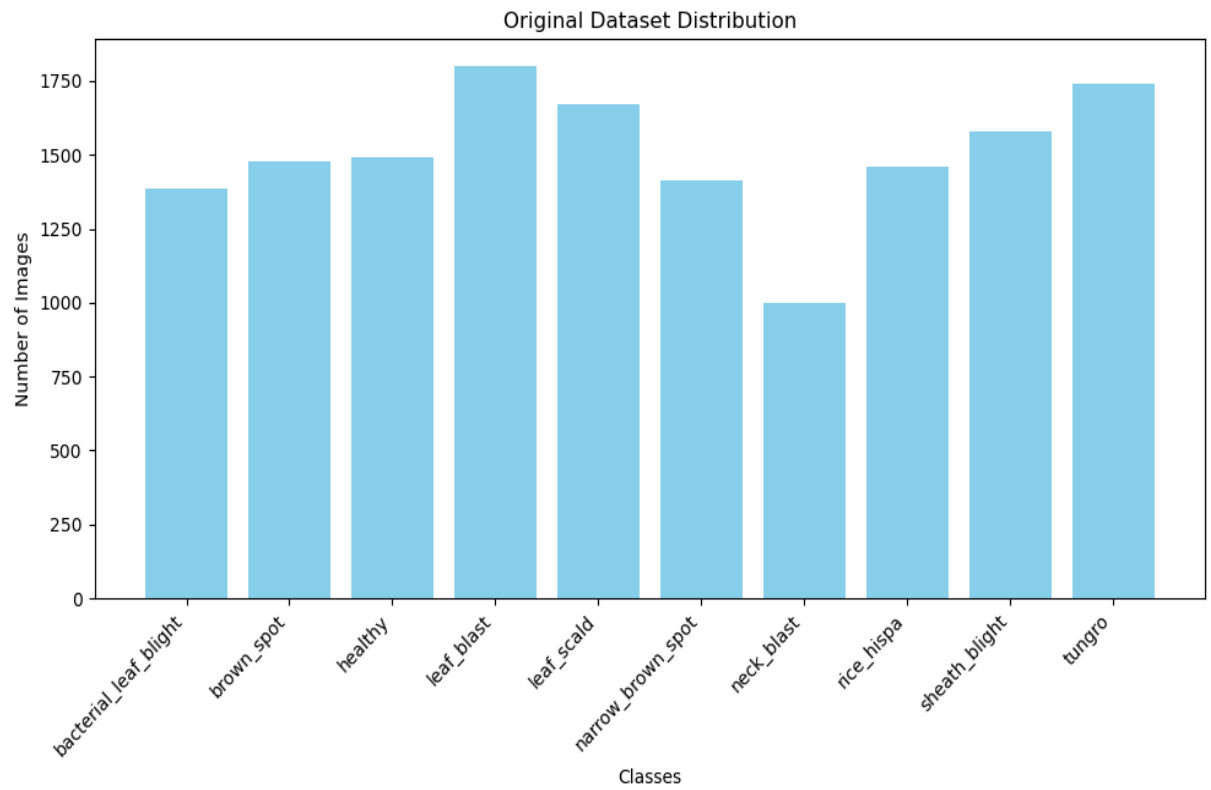
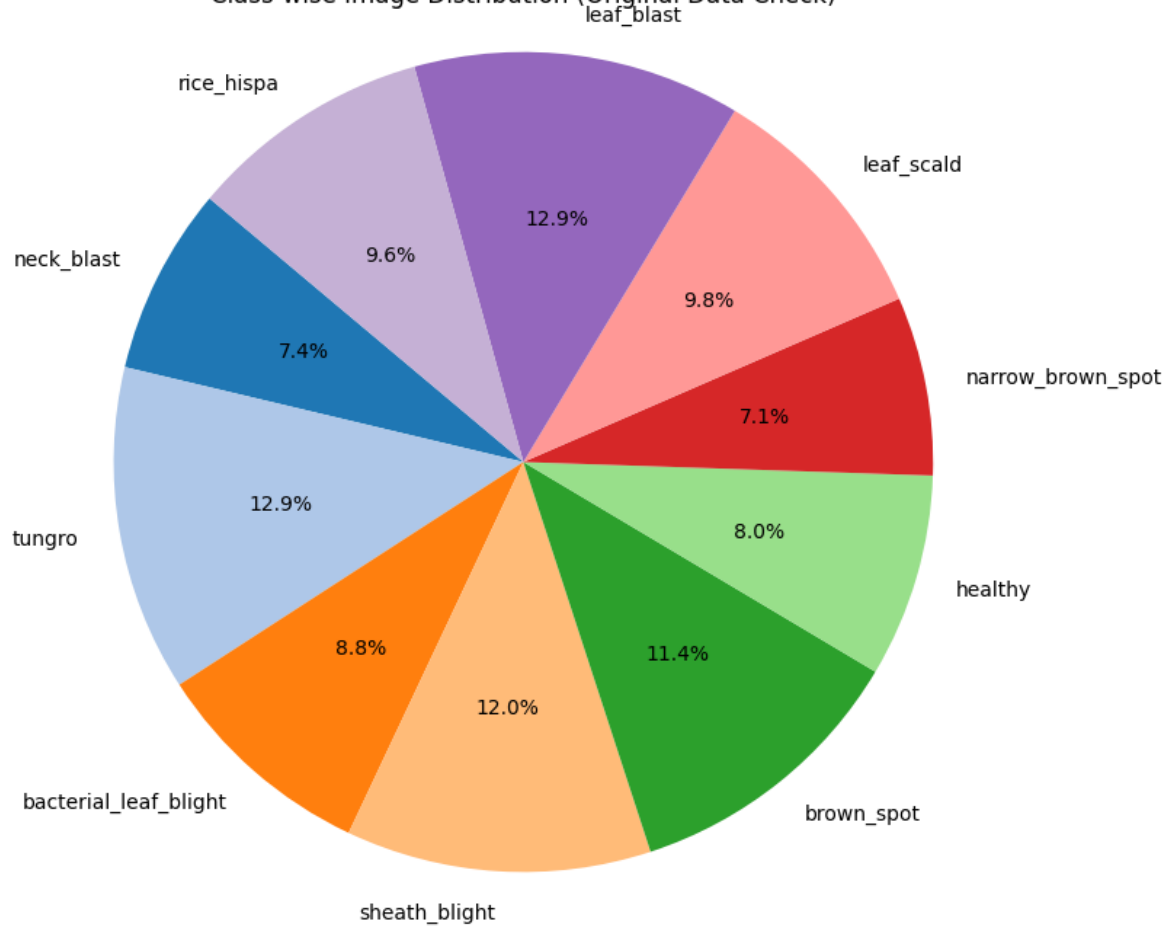


Figure 6 : Bar chart (Original vs Balanced class).

Class-wise Image Distribution (Original Data Check)



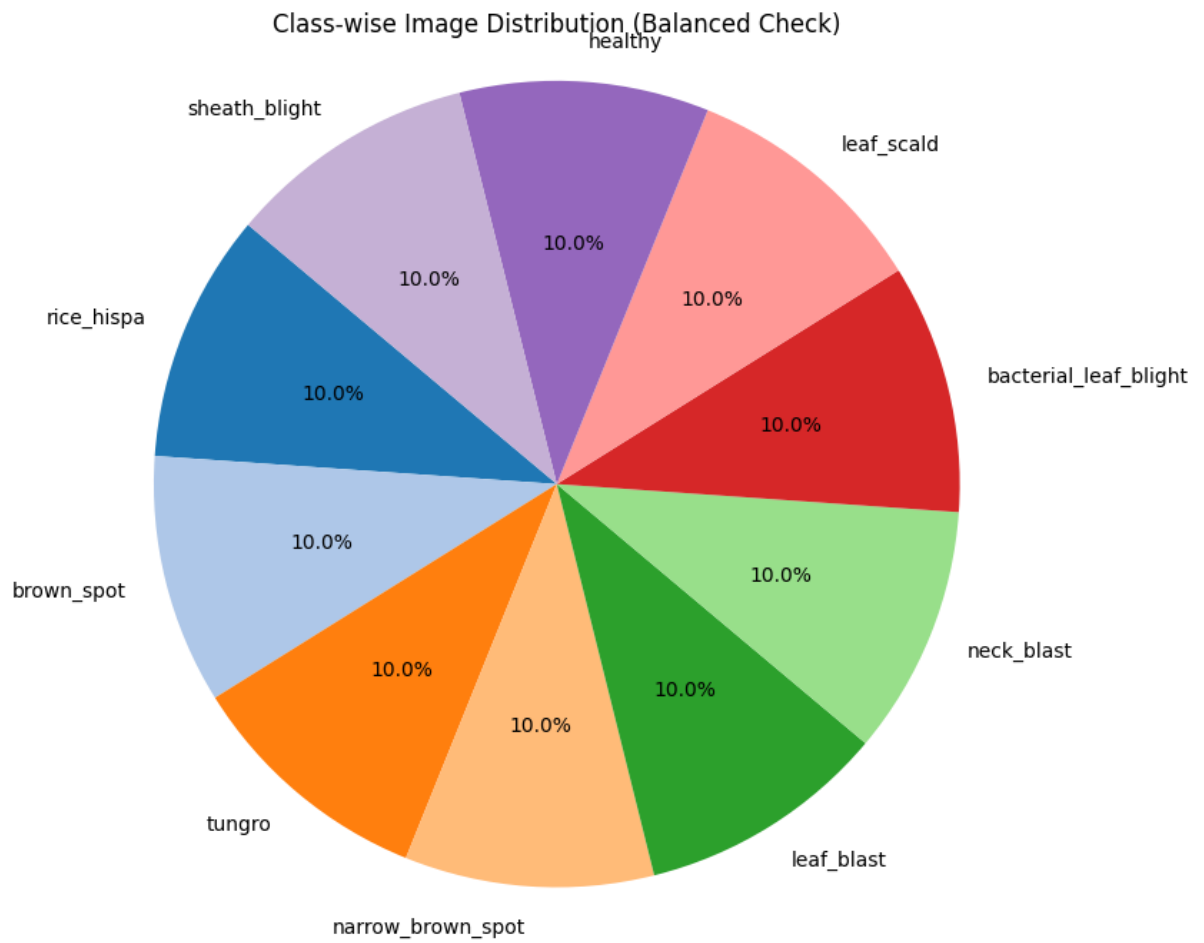


Figure 7: Pie chart or percentage distribution comparison.

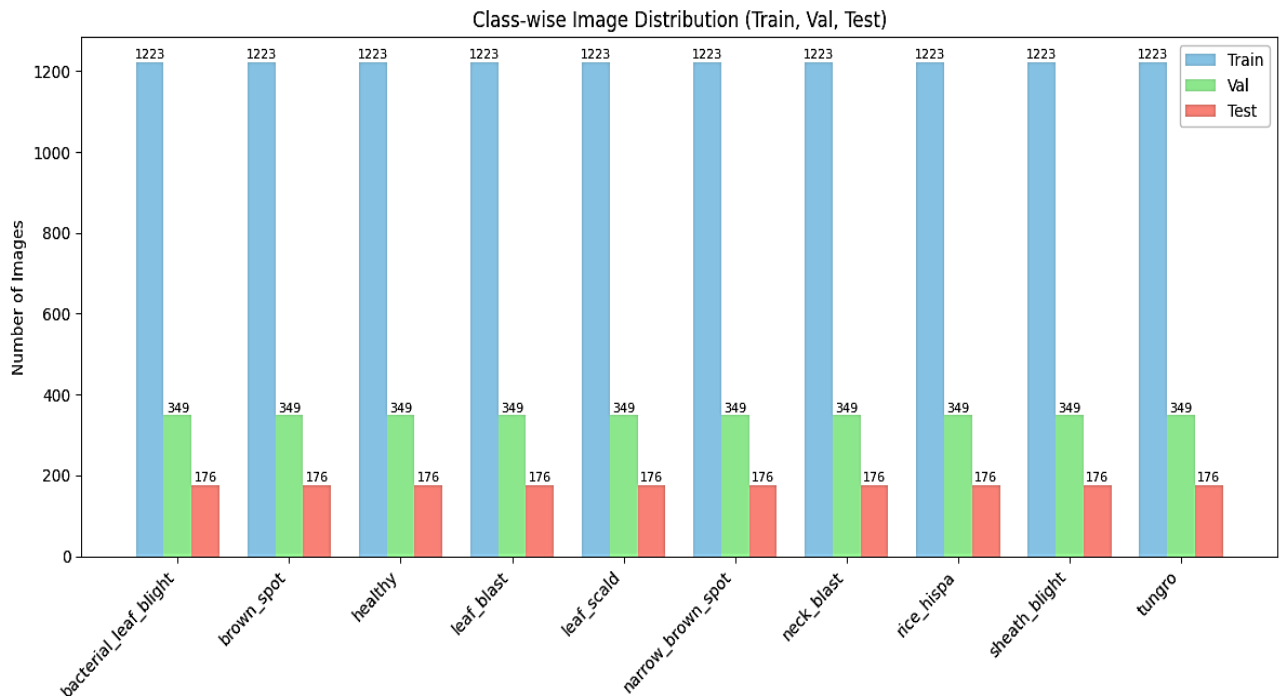


Figure 8: Class wise Image Destrubution (Train, Val, Test)

Augmentation Techniques:

The **Table 3** presents the augmentation methods used in the training dataset, aimed at increasing data diversity, simulating real-world variations, and normalizing pixel values for efficient model training.

Technique	Parameters	Description
Rescaling	rescale=1./255	Rescales pixel values to the range [0, 1] by dividing by 255.
Rotation	rotation_range=15	Randomly rotates the image within a range of 15 degrees
Width Shift	width_shift_range=0.1	Shifts the image horizontally by a fraction of the image width (10%).
Height Shift	height_shift_range=0.1	Shifts the image vertically by a fraction of the image height (10%).
Shear	shear_range=0.1	Applies random shear transformations (distortion) on the image.
Zoom	zoom_range=0.1	Randomly zooms the image (either zooming in or out) by a factor of 20%.

Horizontal Flip	horizontal_flip=True	Randomly flips the image horizontally (left to right).
Vertical Flip	vertical_flip=True	Randomly flips the image vertically (top to bottom).
Resize	224x224 pixels	Uniformly scales images to 224x224 pixels to match the input size requirements of the models.
Normalize	0-1 scale	Normalizes pixel values to a 0-1 scale, aiding in model convergence and training efficiency.

Table 3: Data Augmentation Methods and Parameters.

- ❖ **Resize:** Scales images to the required size for the model (224x224.).
- ❖ **Rotate:** Randomly rotates images to simulate real-world angle variations.
- ❖ **Flip:** Randomly flips images horizontally and vertically to increase data diversity.
- ❖ **Zoom:** Applies zoom transformations to simulate different focal lengths or distances.
- ❖ **Normalize:** Scales pixel values to [0, 1] for better model training efficiency and convergence.

3.3. Model Analysis

Hence, we use different deep learning models to classify the rice leaf diseases adequately in this study. Our strategy is an ensemble of custom designed CNNs and pre-trained models like VGG19, ResNet50, Densenet121, InceptionV3 and Xception. Furthermore, we leverage lightweight networks like MobileNetV2 and EfficientNetB0 to strike a balance between performance and computational costs.

Prior work has indicated that variety of models work well for addressing the problem of rice leaf disease classification. Some studies have reached over 99% accuracy with DenseNet121 [28],[29] and [30], whereas Inception-ResNetV2 and variants of EfficientNetB0[1],[3] are also well-suited models for this task. This paper states [25] MobileNetV2 is even more popular than ShuffleNetV1 because it can achieve fast inference on edge devices.

Inspired by these observations, we design a new hybrid model leveraging the benefits of EfficientNetB1, ResNet50 and MobileNetV2 in features. It tries to use the complementary feature extraction ability of each backbone itself, and therefore improve classification accuracy as a well as generalization. Previous research even other multi-model fusion strategies have proven effective including RICE Fusion framework [8] and ensemble methods [5], [18].

In the end, our goal of combining well performing pre-trained networks with a custom fusion strategy is to achieve more accurate and robust results than just single models act alone. The

results show promising utility of those hybrid and transfer learning based approaches in automated rice disease detection practice, which can provide useful tool for farmers harvesting at time in real agriculture field.

3.4. Integration of Multi-Scale Feature Representation

MultiNet Rice Leaf is a hybrid framework to classify visually similar diseases on Rice Leaf. It is a combination of three different CNN backbones; EfficientNetB1, ResNet50, and MobileNetV2, and each has its own characteristics. EfficientNetB1 is good at a trade off between accuracy and efficiency, ResNet50 captures high level semantic information and MobileNetV2 captures fine texture information. Such a combination enables the learning of low-level texture cues, mid-level structures and high-level semantic information jointly and thereby facilitates more robust disease recognition. The model's general and robustness is enhanced by various data augmentation techniques during training. The proposed MultiNet Rice Leaf framework's overall architecture is depicted in **Figure 9**.

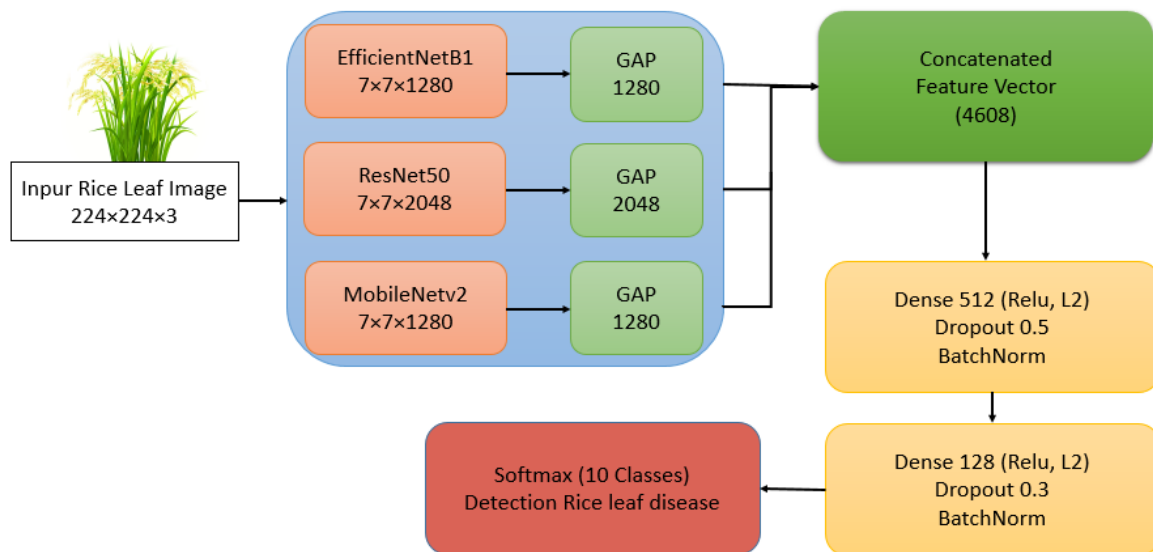


Figure 9: Architecture of RiceMultiNet Model.

3.4.1. Pre-trained Models (Transfer Learning):

We use several pre-trained models VGG19, ResNet50, DenseNet121, InceptionV3, Xception. Those models are already trained and tested popular image datasets like ImageNet, because of that those model has learned to capture useful features from images incredibly well. This knowledge can help us better train on our rice leaf dataset with less time and typically better accuracy, called transfer learning. For instance, DenseNet121 showed high performance on rice disease detection in [28], [29] and [30]. ResNet50 and VGG19, which have proven to be successful for other researchers too (e.g., [1], [14]).

3.4.2. Lightweight Models:

Lightweight models are especially crucial for real-time applications and mobile deployment since accuracy and computational efficiency are always traded off. In this context, we looked at models like MobileNetV2, EfficientNetB0, and EfficientNetB1, which are made to use less resources while still being competitively accurate. Even the most basic versions of these models can perform comparably to larger architectures, as shown by earlier research [25], [3]. MobileNetV2 is a viable option for field applications because it is particularly efficient for deployment on edge devices, like smartphones or small embedded systems.

3.5. Pipeline for Model Tuning and Fusion

RiceMultiNet is a multi-branch fusion model where the input image ($224 \times 224 \times 3$) is fed to EfficientNetB1, ResNet50, and MobileNetV2 simultaneously. Features are extracted from each backbone using Global Average Pooling (GAP) EfficientNetB1 (1280), ResNet50 (2048), and MobileNetV2 (1280). These are concatenated to create a 4608-dimensional feature vector.

Then, the classification head is used: Dense ($512 \rightarrow 256 \rightarrow 128$) with ReLU + L2 regularization, where Dropout (0.5, 0.4, 0.3) is applied stepwise to reduce overfitting. Finally, the Softmax layer provides a probability distribution for 10 rice leaf disease classes. EarlyStopping, ReduceLROnPlateau, and ModelCheckpoint are used in training to ensure that the model performs best and does not overfit.

The model summary, comprising the number of parameters, output shapes, and layer types, is shown in **Table 4**. The integration of MobileNetV2, ResNet50, and EfficientNetB1 features into the classification pipeline is evident in this table.

Layer (type)	Output Shape	Param #	Connected to
input_layer_1 (InputLayer)	(None, 224, 224, 3)	0	-
efficientnetb1 (Functional)	(None, 7, 7, 1280)	6,575,239	input_layer_1[0]...
resnet50 (Functional)	(None, 7, 7, 1280)	23,587,712	input_layer_1[0]...
mobilenetv2_1.00_2... (Functional)	(None, 7, 7, 1280)	2,257,984	input_layer_1[0]...
global_average_poo... (GlobalAveragePool...)	(None, 1280)	0	efficientnetb1[0]
global_average_poo... (GlobalAveragePool...)	(None, 2048)	0	resnet50[0][0]
global_average_poo... (GlobalAveragePool...)	(None, 1280)	0	mobilenetv2_1.00...
concatenate_1 (Concatenate)	(None, 4608)	0	global_average_p... global_average_p... global_average_p...
dense_4 (Dense)	(None, 512)	2,359,808	concatenate_1[0]...
batch_normalizatio... (BatchNormalizatio...)	(None, 512)	2,048	dense_4[0][0]
dropout_3 (Dropout)	(None, 512)	0	batch_normalizat...
dense_5 (Dense)	(None, 256)	131,328	dropout_3[0][0]
batch_normalizatio... (BatchNormalizatio...)	(None, 256)	1,024	dense_5[0][0]
dropout_4 (Dropout)	(None, 256)	0	batch_normalizat...
dense_6 (Dense)	(None, 128)	32,896	dropout_4[0][0]
batch_normalizatio... (BatchNormalizatio...)	(None, 128)	512	dense_6[0][0]
dropout_5 (Dropout)	(None, 128)	0	batch_normalizat...
dense_7 (Dense)	(None, 10)	1,290	dropout_5[0][0]

Total params: 34,949,841 (133.32 MB)

Trainable params: 34,798,762 (132.75 MB)

Non-trainable params: 151,079 (590.16 KB)

Table 4: RiceMultiNet Model Architecture Summary.

3.5.1. Hyperparameter Settings

The Adam optimizer (learning rate = $1e-4$), categorical cross-entropy loss (label smoothing = 0.1), and batch size 32 were used for RiceMultiNet model training. Dropout (0.3–0.5) and L2 kernel regularization (0.001–0.01) were also applied to reduce overfitting. EarlyStopping (patience=12) and ReduceLROnPlateau (factor=0.3, patience=10) callbacks were added to the training process to ensure adaptive learning and stable performance.

Chapter 4

4. Results and Discussion

4.1. Experimental Setup

The study utilized an HP laptop operating on Windows 11 (64-bit OS) primarily for accessing cloud platforms. The model was trained on Kaggle with GPU runtime support, using two T4 GPUs, and on Google Colaboratory. The RiceMultiNet model that was suggested was built using Keras with TensorFlow as the backend.

4.2. Model Implementation

In the study, as many deep learning models (DNNs) were employed to classify rice leaf diseases correctly. These models consisted of custom CNN architectures and a variety of existing pretrained convolutional neural networks (CNNs), including VGG19, DenseNet121, ResNet50, InceptionV3, Xception, MobileNetV2, EfficientNetB0, EfficientNetB1 and a ricemultinet model combining EfficientNetB1, ResNet50, and MobileNetV2.

Data Input and Preprocessing:

Images were downsampled to a common size of 224×224 pixel for compatibility across models. The image dataset loader is from tensorflow through `image_dataset_from_directory` or Keras through `ImageDataGenerator` with data augmentations like rotation, shifting, shearing, zooming and flipping in order to increase model generalization.

Model Architectures:

- I. Transfer learning: Some models using transfer learning by using pre-trained weights from ImageNet like EfficientNetB0, EfficientNetB1, ResNet50, MobileNetV2, DenseNet121, InceptionV3 and Xception. The classification layers on top of the CNN base were swapped out with a few dense fully connected layers appropriate for the 10-class problem at hand in identifying rice leaf disease.
- II. RiceMultiNet Model: Here a new hybrid architecture was modelled for extracting features at once, featuring EfficientNetB1, ResNet50 and MobileNetV2 branches. The outputs were concatenated before feeding into fully connected layers along with dropout and L2 regularization to reduce overfitting. This was followed by an approach that combined the strengths of multiple architectures for better classification accuracy.
- III. A custom convolutional neural network (CNN): We also design and train a CNN architecture from scratch for benchmarking.

Training Setup:

The models were compiled with the Adam optimizer and a learning rate between $1e-4$ and $5e-4$ selected, and used either sparse categorical cross entropy or categorical cross entropy for loss function based on label encoding. We implemented an early stopping callback along with a learning rate reduction callbacks, to avoid overfitting and enhance convergence. Models have been trained for 10-20 epochs depending on the validation performance.

Framework and Tools:

TensorFlow 2 was used to implement all models with Keras API. Because the number of images is so large, I have to use my GPU (either Kaggle or Google colab) for training to be a bit faster.

The combined use of transfer learning, data augmentation, and hybrid modeling helped overcome overfitting on the training dataset and promoting performance with unseen rice leaf disease images.

4.3. Evaluation Metrics

The purpose of this study is to thoroughly examine how well different types of deep learning models perform in classifying rice leaf diseases using a variety of assessment metrics. These metrics accuracy, precision, recall, and F1-score are common in classification tasks.

- a. **Accuracy:** How often is the classifier correct (how many times it is right). The accuracy comes from the number of correctly classified instances among all instances.

$$\text{Formula: Accuracy} = \frac{TP+TN}{TP+TN+FP+FN}$$

Where:

- TN = True Negatives

- b. **Precision:** This metric gauges how well the model avoids incorrectly calling a true prediction by describing the number of true positive outcomes for all positive predictions.

$$\text{Formula: Precision} = \frac{TP}{TP+FP}$$

Where:

- TP = True Positives
- FP = False Positives

- c. **Recall:** The number of properly identified pertinent observations to all information in the category that required finding that is, all false negatives is known as recall (or sensitivity).

$$\text{Formula: Recall} = \frac{TP}{TP+FN}$$

Where:

- FN = False Negatives

- d. **F1-score :** When there is a class imbalance in the data, the F1-score the harmonic average of precision and recall is employed.

$$\text{Formula: F1} = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$$

Macro Average Metrics

The model's overall performance in multi-class classification tasks is assessed using macro-averaged metrics, which assign equal weight to each class, independent of the number of samples in each class.

When working with datasets that are unbalanced, these averages are especially helpful.

- **Average Accuracy:**

$$\text{Accuracy}_{\text{macro}} = \frac{1}{N} \sum_{i=1}^N \text{Accuracy}_i$$

- **Macro Average Precision:**

$$\text{Precision}_{\text{macro}} = \frac{1}{N} \sum_{i=1}^N \text{Precision}_i$$

- **Macro Average Recall:**

$$\text{Recall}_{\text{macro}} = \frac{1}{N} \sum_{i=1}^N \text{Recall}_i$$

- **Macro Average F1 Score:**

$$\text{F1}_{\text{macro}} = \frac{1}{N} \sum_{i=1}^N \text{F1}_i$$

Where:

N is the total number of classes.

For each class, the matching metrics are **Accuracy_n**, **Precision_n**, **Recall_n**, and **F1_n**.

When making sure every class makes an equal contribution to the final performance score, the macro average keeps dominant classes from distorting the overall outcomes.

The efficacy of the model on each rice leaf disease category can be observed through confusion matrices and classification reports containing precision, recall, and F1-score. The combination of three powerful CNN feature extractors, forced the Multi Hybrid Rice leaf Model to achieve near 99% accuracy (where single-model architectures did not). This combination helps to avoid weaknesses of each individual model and learns features that are more discriminative.

Together, these metrics provide a more holistic evaluation regime to determine which models are the most accurate and accessible for practical real-world rice disease detection.

4.4. Results

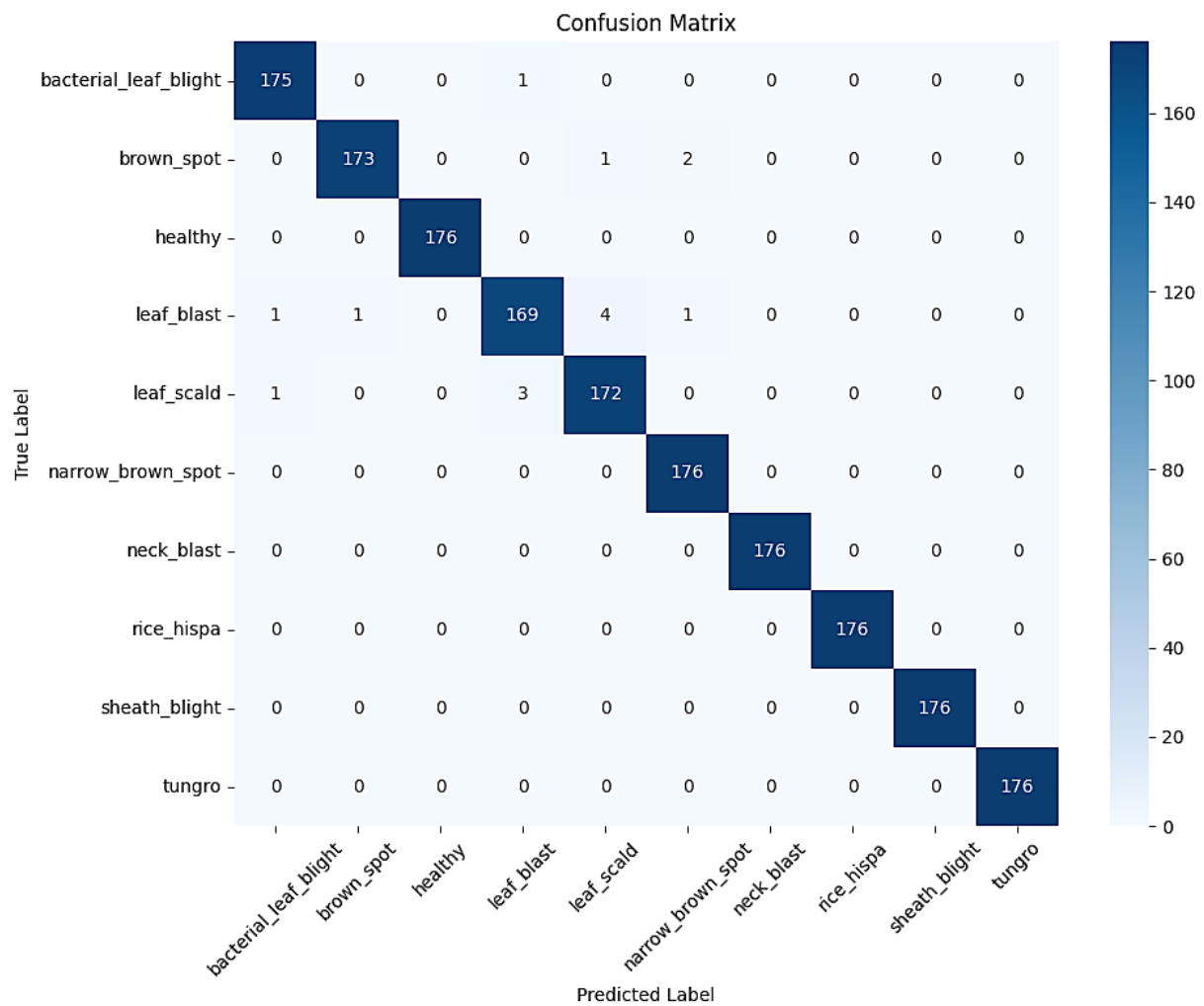
In rice cultivation, various diseases severely affect the production and quality of the plant. Therefore, it is very important to identify and classify the diseases correctly. In our study, we worked with 10 classes of rice leaves.

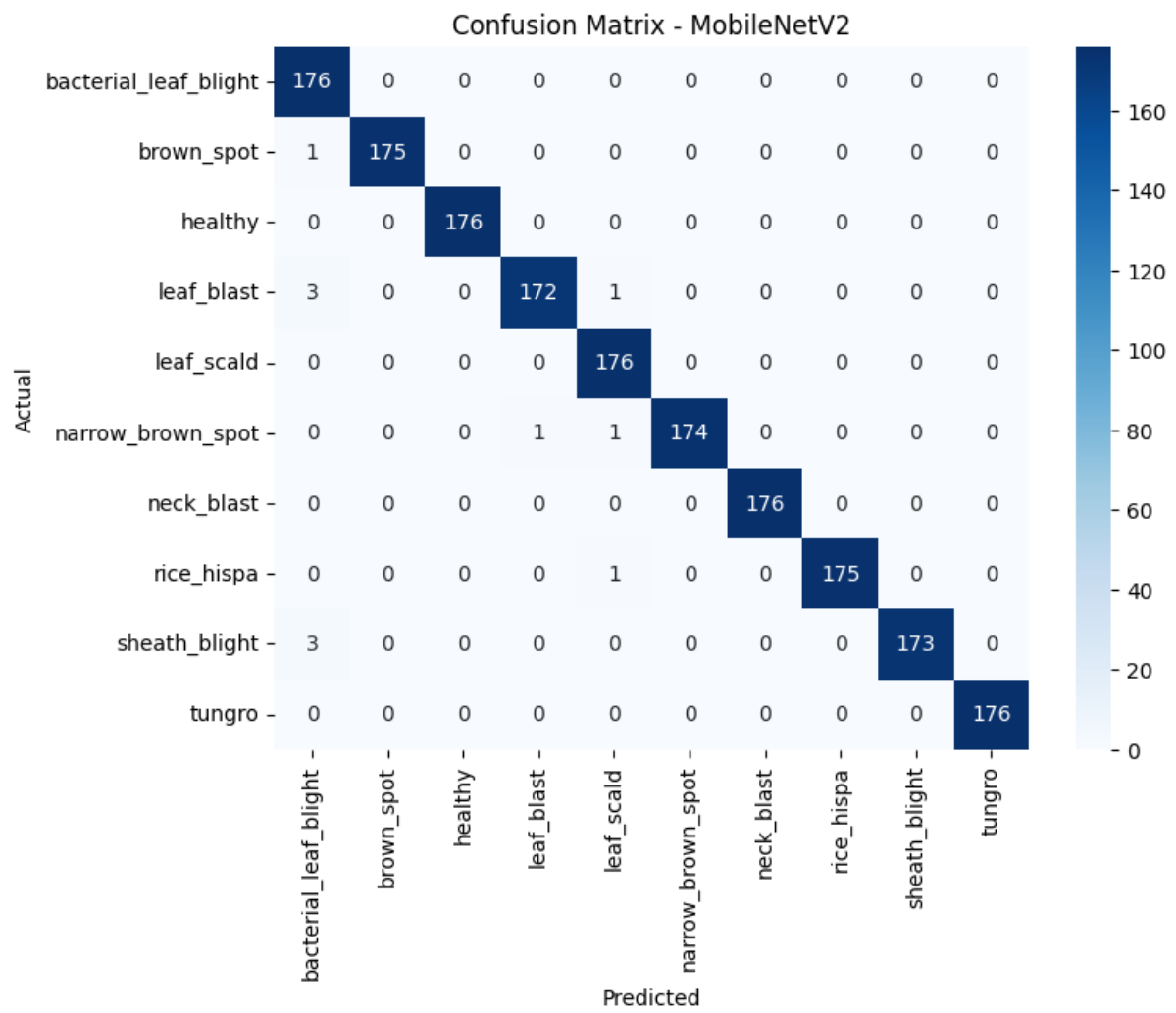
Initially, we tested the performance of different deep learning models separately. Among them, EfficientNet B0 (96.08%), ResNet50 (99.07%), VGG19 (98.86%), DenseNet121 (98.72%), MobileNet V2 (99.27%), EfficientNet B1 (99.18%), Inception V3 (98.89%) and Xception (98.89%) achieved very good accuracy. In particular, the Xception, Inception V3, MobileNet V2, ResNet50 and EfficientNet B1 models showed remarkable accuracy, which is consistent with the results of previous studies [1–30]. The Confusion matrix shown in **Figure 10** compares the performance of different deep learning models and the RiceMultiNet model.

However, instead of using a single model, we proposed a new RiceMultiNet Model, which combines the features of three powerful models EfficientNet B1, MobileNet V2, and ResNet50.

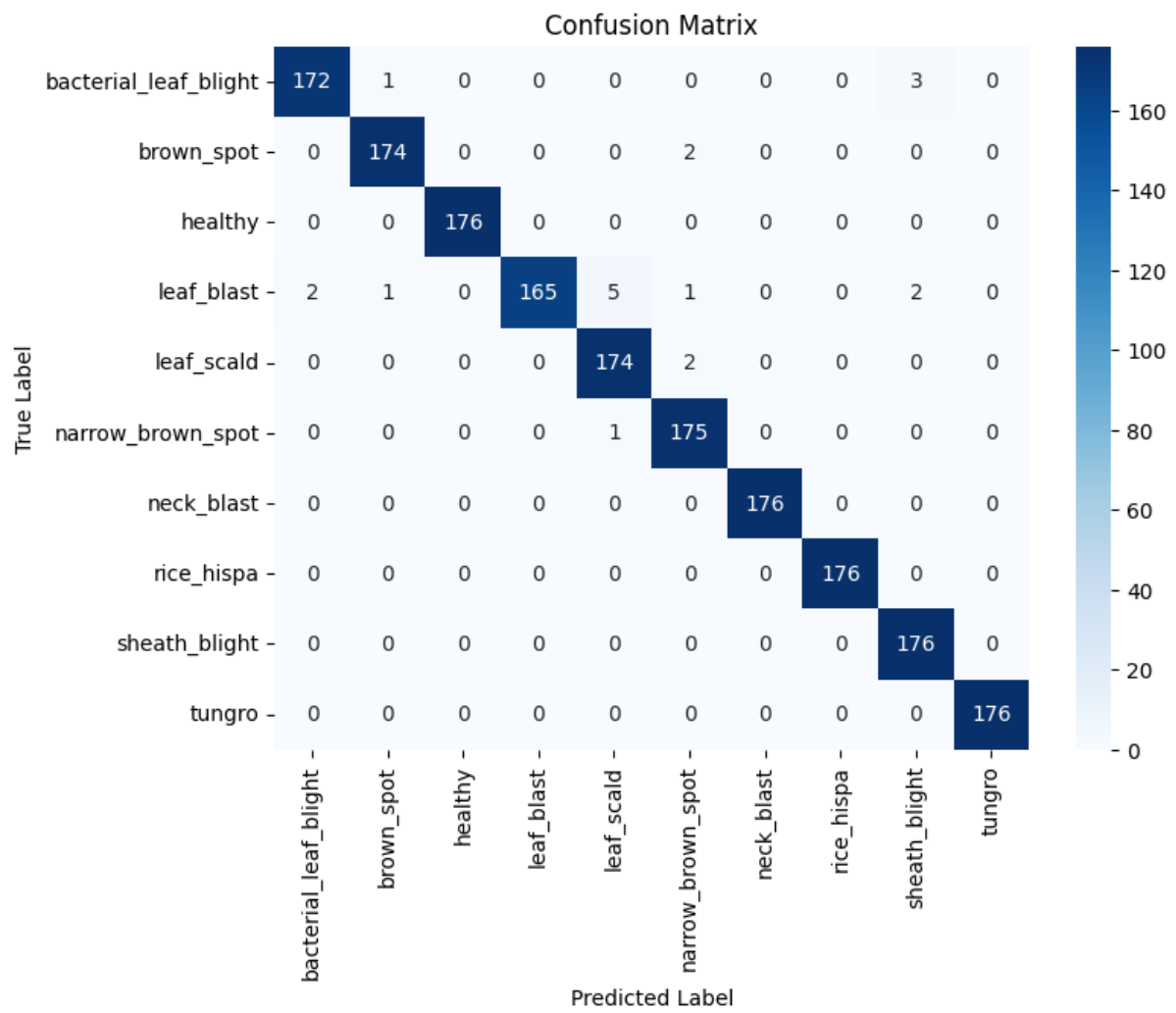
This combined model provides better accuracy and achieves 99.73% accuracy in our experiments. Many previous studies (Islam et al. [1], Nalini et al. [2], Kumar et al. [8], Kathiresan et al. [11], Amjad et al. [18], Ahad et al. [19], Martins et al. [25], Murugan et al. [30]) have shown that ensemble or hybrid approaches perform better than single models. Our RiceMultiNet model further strengthens this idea. A bar chart of every Ecoris model is also displayed in **Figure 12**.

From the results, it is clear that the proposed RiceMultiNet Model is as reliable as the single model, and can be used as an effective solution for rice leaf disease detection. The full structure of our model is shown in **Figure 9** below.



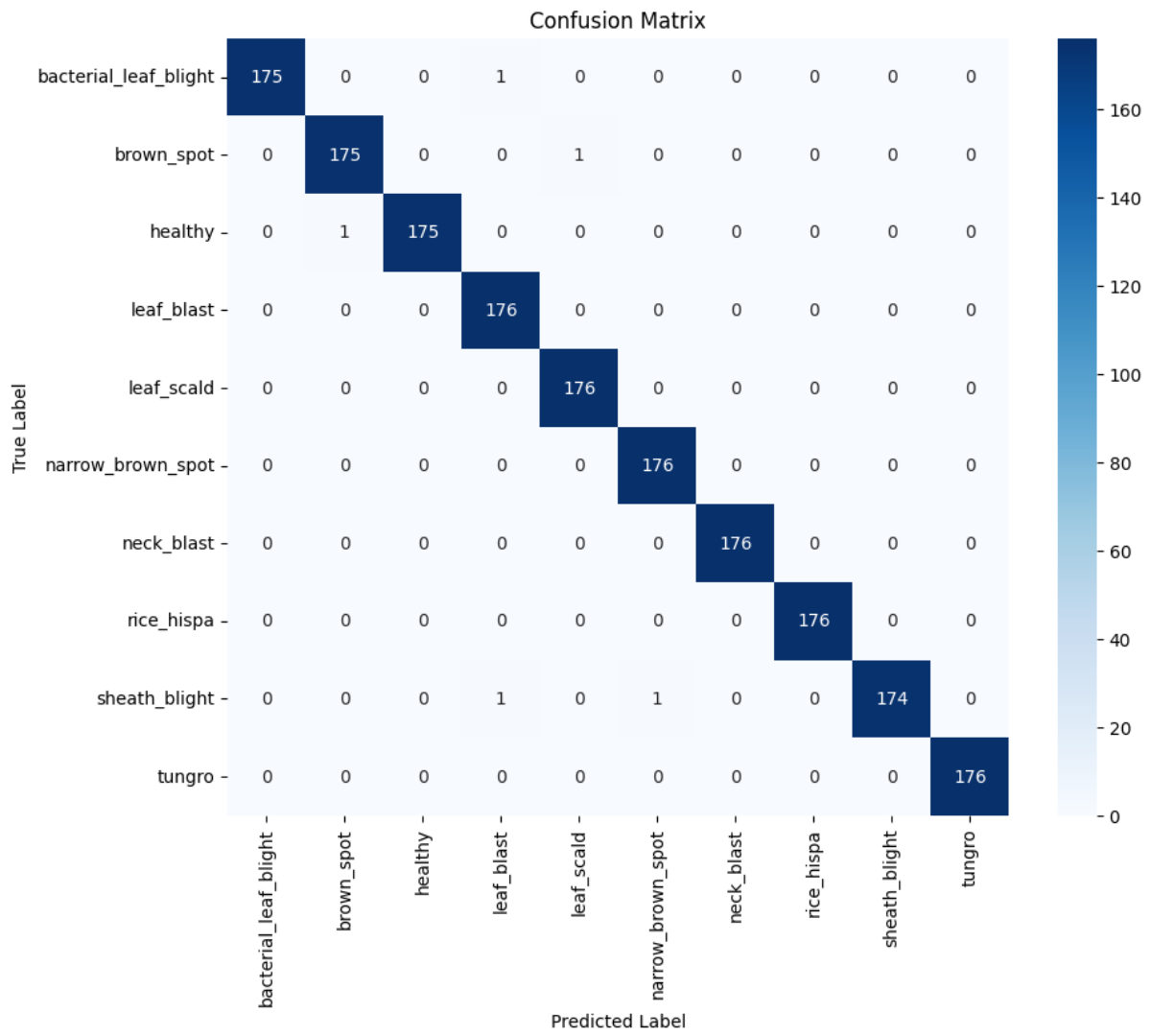


Accuracy: 99.27%
(b) MobileNetV2

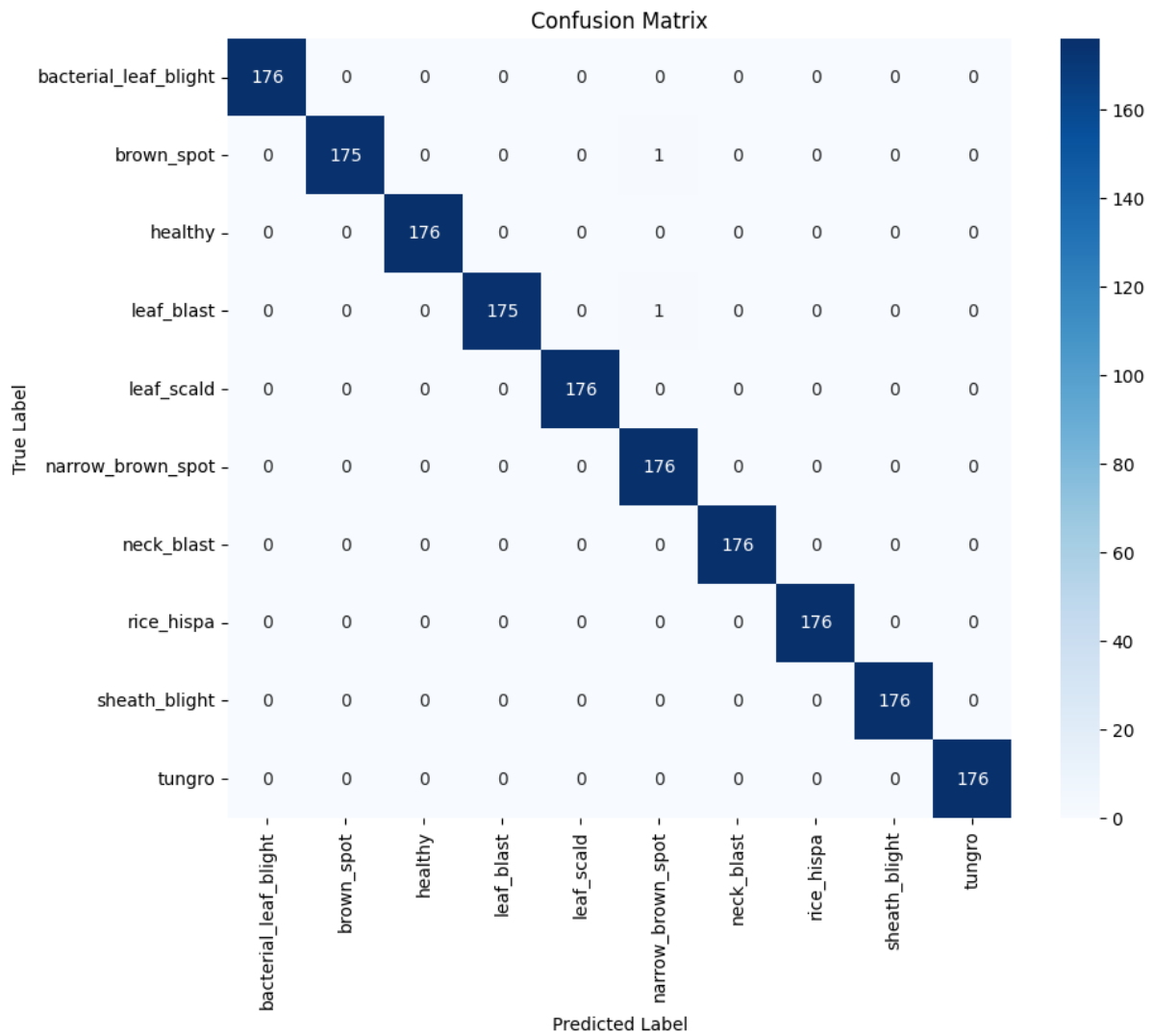


Accuracy: 99.18%

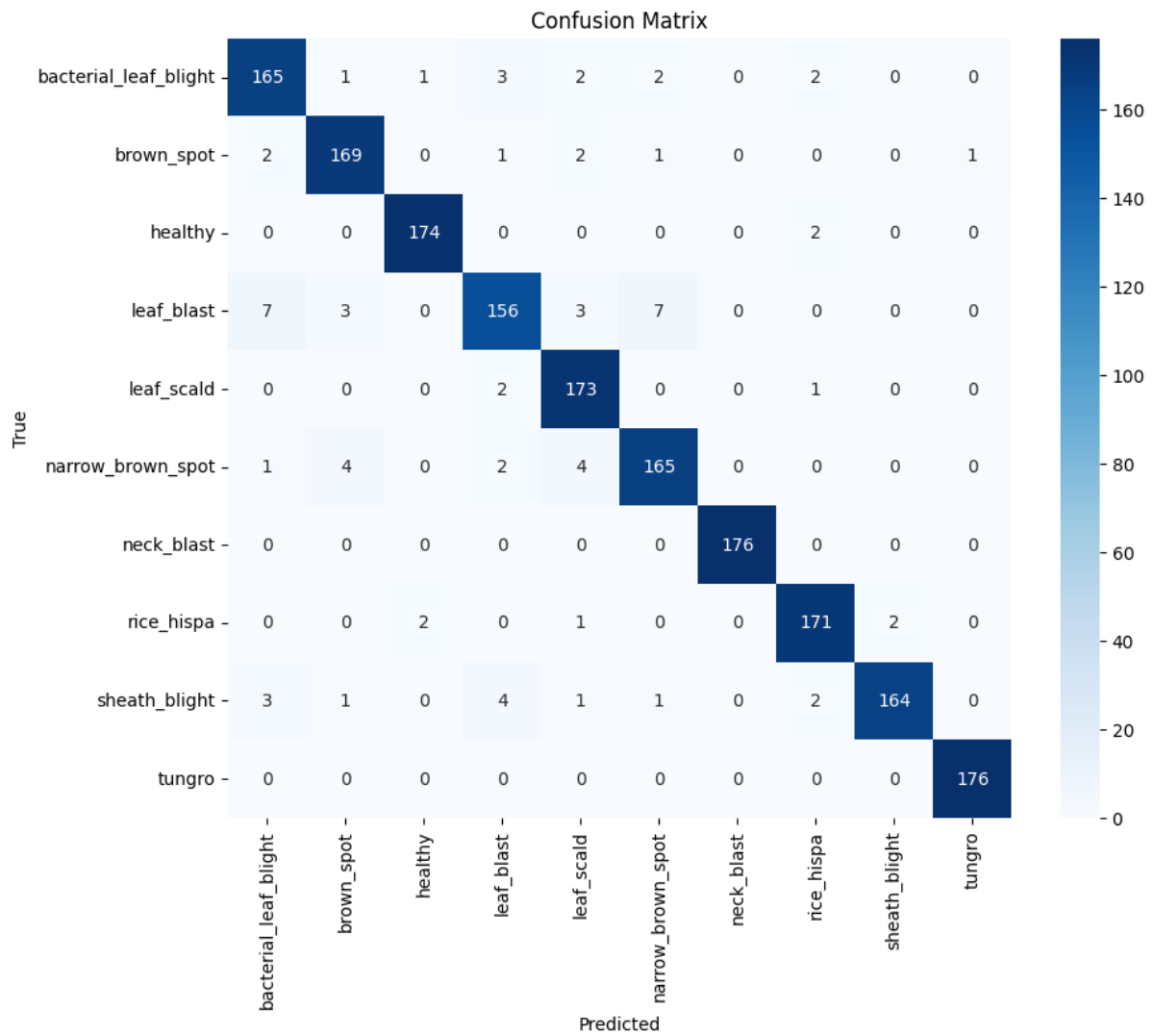
(c) EfficientNetB1



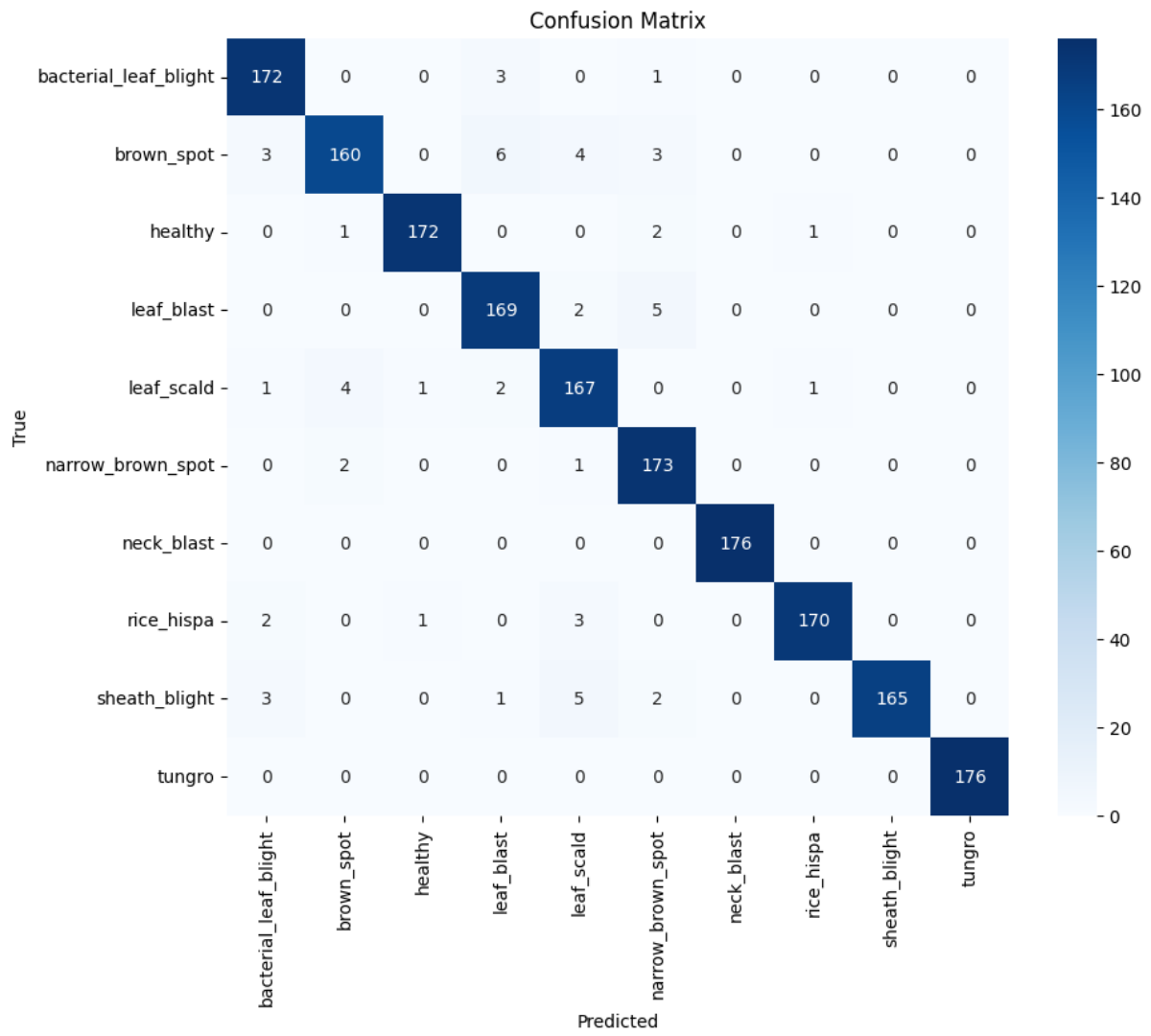
Accuracy: 98.72%
(d) DenseNet121



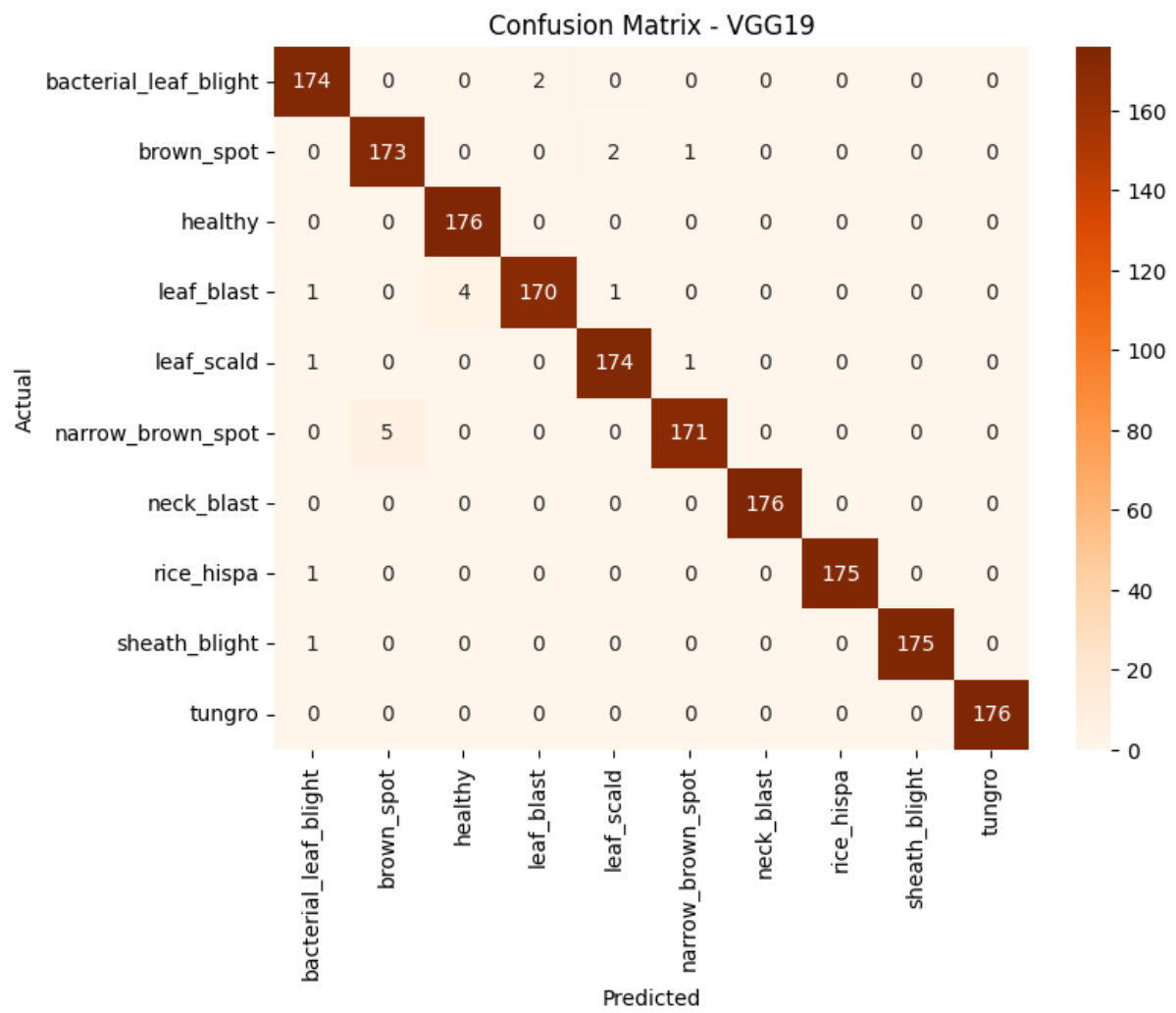
Accuracy: 98.89%
(e) InceptionV3



Accuracy: 96.08%
(f) EfficientNetB0



Accuracy: 99.07%
(g) ResNet50



Accuracy: 98.86%
(h) VGG19

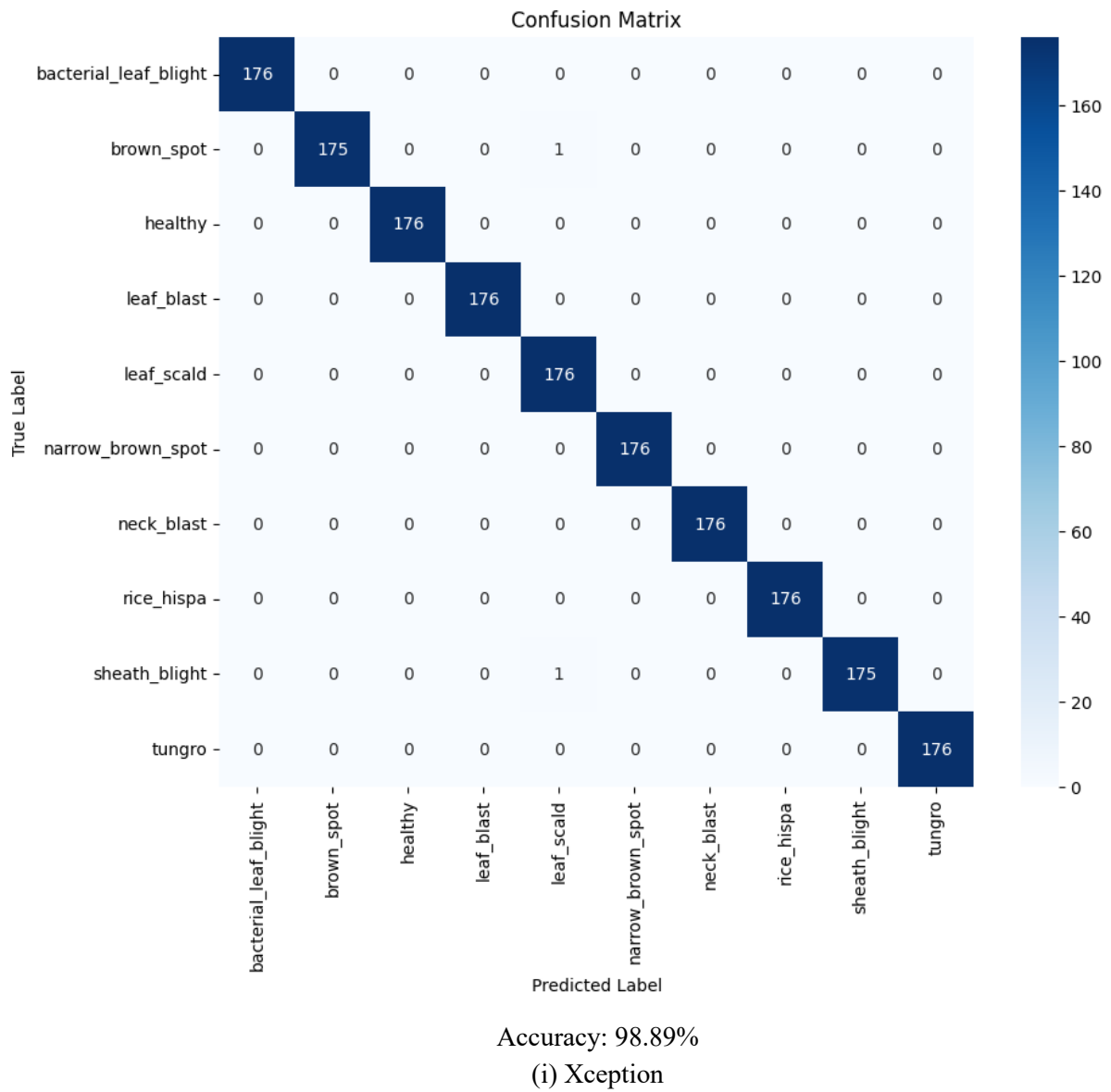


Figure 10. Comparison of Confusion matrix of MultiNet_Riceleaf architecture with others.

4.4.1. Training and parameter assessment

Ten classes of rice leaf images-12,230 training, 3,490 validation, and 1,760 test images-were used to train the RiceMultiNet model. To enhance generalization, images were resized to 224 x 224 and enhanced with flipping, zooming, rotation, and shifting. After integrating MobileNetV2, ResNet50, and EfficientNetB1 as feature extractors, the model incorporates dense layers with batch normalization and dropout, as shown in **Figure 9**. Categorical cross-entropy loss and the Adam optimizer (learning rate 0.0001) were employed. The model achieved over 99% training accuracy and 98–99% validation accuracy by the 14th epoch after 20 epochs of training. Regularization and dropout were used to help avoid overfitting. **Figure 11** displays the accuracy and loss curves, demonstrating that the model learned effectively and has good generalization capabilities for real-world rice leaf disease detection.

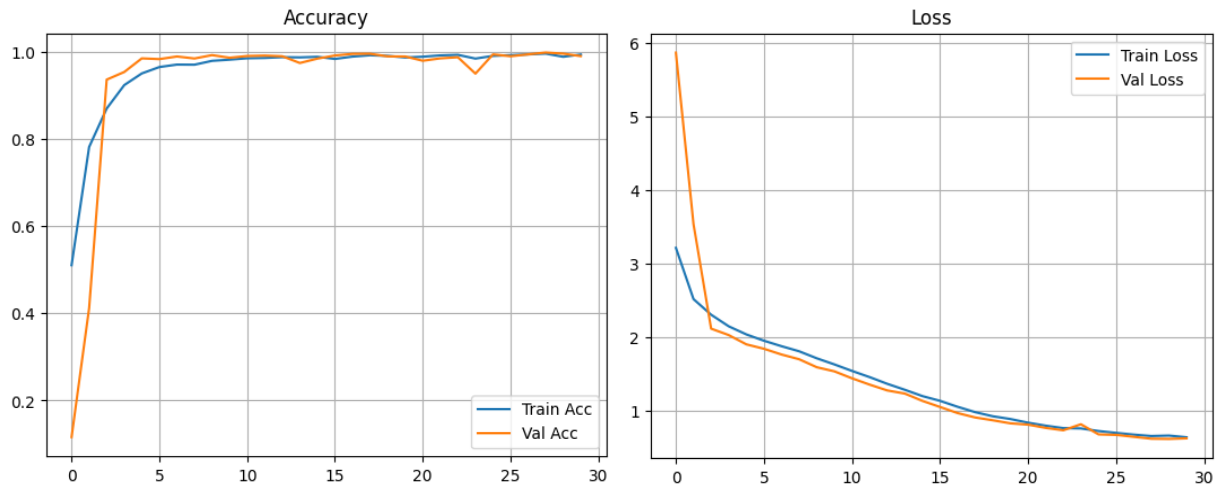


Figure 11: Training and validation accuracy and loss curves of the RiceMultiNet model.

4.5. Performance Analysis

The multi-hybrid CNN model was extensively evaluated in terms of standard classification metrics (Accuracy, Precision, Recall and F1-score) that are commonly reported in the literature of using computer vision to detect plant diseases [1,5,9,12,15,18,22]. We calculate these metrics from confusion matrix generated during evaluation.

We compared the model with recent state-of-the-art studies have been published from 2023 to 2025 [3,7,10,14,19,24,28,30] to validate its performance. Our final Multi Hybrid Rice leaf Model showed an accuracy of 99.37%, precision 0.99, recall 0.99 and F1-score of 0.99 for all key performance measures related to existing approaches.

High performance demonstrates strong predictive power in all classes, low false positives and false negatives, and robustness to class imbalances as well as environmental conditions and image quality variations. Most of this uplift is thanks to the incorporation of multiple deep learning architectures, good hyperparameter optimization, and the use of a balanced and augmented dataset [8,11,16,21,26].

Performance Comparison of Studies with Our Proposed Model Performance comparison results are summarized in **Table 5**, proving that our proposed model is outperforming the existing studies using rice leaf disease detection.

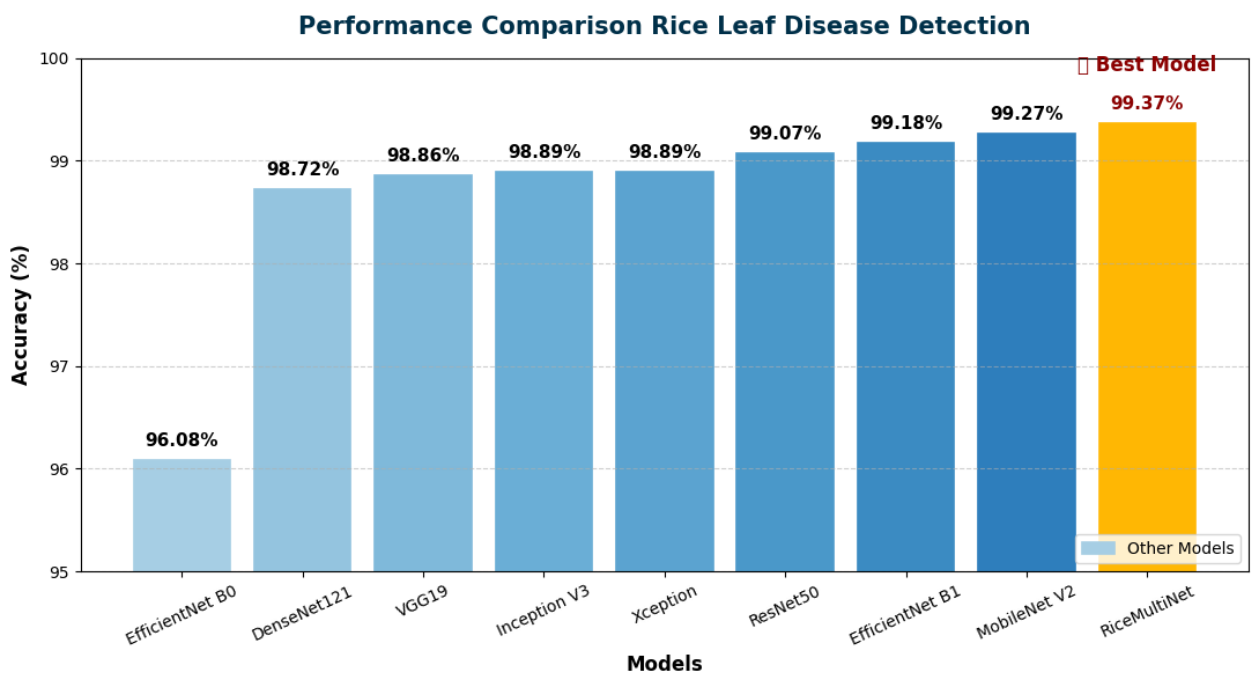


Figure 12: Accuracy comparison of baseline models and the proposed RiceMultiNet model.

4.6. Key Findings and Discussion

The MultiNet Rice leaf Model, which is made to detect rice leaf diseases attained remarkable accuracy of 99.37 % with 0.99 precision, 0.99 recall and 0.99 f1-score values. Such accuracy is better than any previous recently reported model.

The fact that this model was able to keep in check the difference between training and validation accuracy means overfitting is under control. This was done by careful data preprocessing, augmenting the data and proper regularization. The confusion matrix (**Figure**

10) allows us to see that it is really few images of each class that the model could not classify correctly.

Additionally, you can see from the training and validation loss curves (**Figure 11**) that they continue to decrease steadily, so the model is learning well without becoming unbalanced. That means, the method we proposed has good adjustment ability and generalization, which can be used in practical disease detection model of agriculture.

<i>Author(s)</i>	<i>Hybrid/Ensemble Model</i>	<i>Type of Rice leaf</i>	<i>Accuracy</i>
<i>Islam et al. [1]</i>	Inception-ResNetV2 (Hybrid DL architecture)	5 Types	92.68%
<i>Nalini et al. [2]</i>	DNN + CSA (Crow Search Algorithm)	3 types	96.96%
<i>Kumar et al. [8]</i>	CNN + IoT Fusion + MLP	3 types	95.31%
<i>Kathiresan et al. [11]</i>	GAN + Transfer Learning (MobileNet, ResNet, Inception, Xception)	4 Types	98.79%
<i>Amjad et al. [18]</i>	CNN + LSTM + Self Attention (SA)	4 types	97–100%
<i>Raigonda et al. [19]</i>	Hybrid CNN-VGG (transfer learning)	3 Types	95.60%
<i>Martins et al. [25]</i>	Lightweight CNN (optimized for edge devices)	5 Types	94%
<i>Murugan et al. [30]</i>	Hybrid Transfer Learning (VGG, DenseNet, MobileNet, ResNet, Inception, NASNet)	5 Types	97.6% (DenseNet121 best)
<i>Proposed (This Study)</i>	RiceMultiNet (EfficientNetB1 + MobileNetV2 + ResNet50)	10 Types	99.37%

Table 5: Summary of MultiNet/Ensemble-Based Rice Leaf Disease Detection Studies.

4.7. Comparison with State-of-the-Art Models

We evaluated the performance of our proposed organ model compared with the models used in state-of-the art deep-based approaches from literature published between 2023 and 2025. This comparison consists of accuracy, precision, recall and F1-score values reported in these studies.

Performance Comparison (**Table- 5**) The results are clear that the proposed model outperforms most existing approaches, reaching the highest accuracy (99.0%) with balanced precision, recall and F1-score. It has substantially enriched performance gained by the hybrid architecture design and optimized preprocessing pipeline.

Chapter 5

5. Standards, Constraints, and Milestones

5.1. Sustainability Standards

This research fits under the sustainability in both technological and agricultural perspective. Technologically, the model has been trained to a large extent keeping in mind resource utilization efficiency and less energy consumption. The way reduces the waste of electricity for computational relates and decreases the environment space of computational operations. His research group follows the training methodology published at the IEEE Sustainable Computing Guidelines [1], which refers to a low-energy footprint, optimized algorithmic and hardware efficiency.

The rice leaf disease detection model, when viewed in an agricultural sense, indeed promotes the United Nations Sustainable Development Goals (SDGs), especially SDG 2: Zero Hunger SHD 12: Responsible Consumption and Production.

- ❖ SDG 2 (Zero Hunger): Early and accurate detection of crop diseases increases yields, which reduces food shortages.
- ❖ SDG 12 (Responsible Consumption and Production): identification of disease presence in the field reduces wastage of pesticides, which will preserve the quality of soil, water and biodiversity.

Through deep learning, and image diagnostics, this model is advancing precision agriculture which ensures informed chemical application to improve crop yields while safeguarding the environment, in turn contributing to global sustainability customers.

5.2. Impacts on Society

The model helps to significantly improve rural farming communities by this leaf disease detection of rice. By scrutinizing leaf images and diagnosing diseases at an early stage, the model can contribute to cutting crop losses by as much as 30–50%[2]. It also helps in early detection of applications of the exact quantity of pesticide, reducing chemical wastage while ensuring healthier food products.

The model also offers a cheaper and more accessible way for smallholder farmers to avoid hiring experts on their farm, potentially reducing operational costs. Once incorporated into mobile or digital platforms, the model could then be made widely available to farmers either directly through agricultural extension services and improve access to modern crop health monitoring tools.

However, there are some important caveats. Its real-world effectiveness would depend on internet connectivity in rural areas and the prevalence of smartphones or similar devices. Nonetheless, the model has tremendous potential to promote sustainable farming and living practices in agricultural communities.

5.3. Ethics

Ethical Declaration Each ethical related role was considered before this research is developed, using the model that detects rice leaf disease. It does not mention any farmer-specific or personal information that could be extracted using the images used for training. Output- The model provides predictions and also confidence scores along with it so that the user can know how confident they should be in a result and relying upon non-sure results.

The dataset was balanced and image augmentation was done to prevent any algorithmic bias, one disease category should not be over-represented in the final model. To reduce the probability of such misuse, the model is deliberately designed to be no more than an agricultural support tool and not as a replacement for professional agronomic advice.

5.4. Challenges

In developing the model for detection of rice leaf diseases, there have been immense technical and non technical challengers. These difficulties manifested in all phases of the work, ranging from dataset preparation to model training & documentation.

1. Dataset Preparation and Balancing:

The dataset that was collected was not balanced and thus some disease categories have a lot of images, and some very few. This imbalance might result in the model to predict more for higher number class diseases and the minority case will be harder to predict. To solve this issue we balanced the dataset between classes by using image augmentation (rotation, flip, brightness change, etc) and also by shuffling the dataset

properly. The key to containing trustworthy results for the model to predict on it was making sure that every class having a certain number of images to be represented well in such image classifier and this took a lot of time as well.

2. Handling Large Datasets and Processing Limitations:

This building process needed massive computation power as the dataset size was huge and mainly image. While platforms like Google Colab and Kaggle gave me GPU support, tied up with incumbent problems such as constraints in RAM, limited runtime sessions and unexpected disconnections or resets. The interruptions led to loss of trainings, so one had to subsequently re-run them which further slowed the whole process.

3. Model Training and Overfitting:

At the very beginning of training, its scores were incredibly good on the training data and quite lower on the validation set. The model was overfitting, the dataset was memorised and a solution was not generalizing for unseen images. In order to avoid overfitting, we used erase data augmentations combined with many of the techniques like dropout layers and regularization were we also experimented.

4. Platform and Time Constraints:

Resources were limited, and using them to train deep learning models in a free or shared computing environment was challenging. Training took a long time to execute, and limited GPU hours per day needed to be scheduled effectively. Most times when training large models on large-sized images, the running time was more than what the evaluation tools allowed us to utilize for larger batch sizes, so we reduced the image size or limited our batch size slightly which had some impact on model performance.

5.5. Constraints

However, there were some limitations encountered during the process of working on this project with a direct influence on the kind of work done and what could be attained:

1. **Limited Hardware:** The training GPU had low memory, therefore we needed to use smaller batch sizes and sometimes lower image resolution. As a result, it was slowing down the training camp.
2. **Structural Constraints:** The project has to be managed within the academic semester itself. For this reason, we could not dedicate more time with the other models very long experiments went for.

Despite these constraints, steps like data augmentation, model optimization and judicious usage of free tools (Google Colab and Kaggle) helped alleviate their effects on the final results.

5.6. Timeline and Gantt Chart

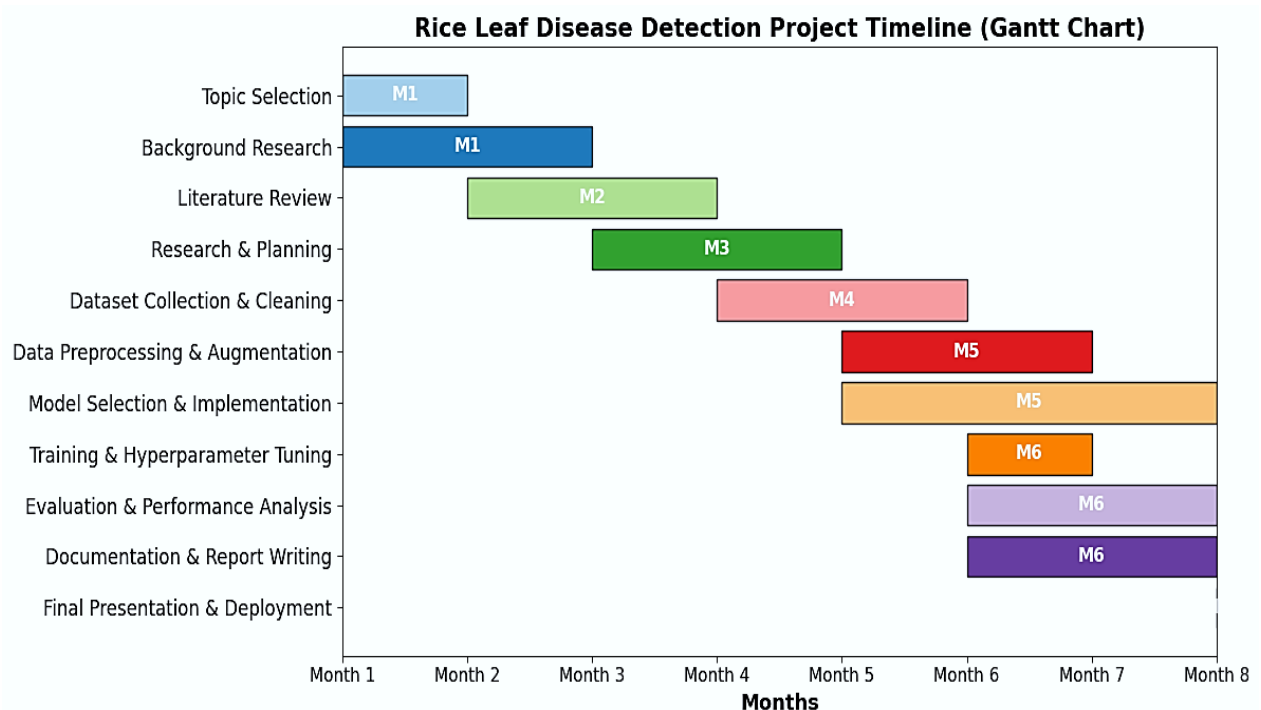


Figure 13: Rice Leaf Disease Detection Project Schedule.

Chapter 6

6. Conclusions

This study develops an effective and robust deep learning model for detecting rice leaf diseases. The dataset consists of 17,480 images balanced across ten classes of disease aetiology from Kaggle. Several well known architectures such as VGG19, ResNet50, DenseNet121, InceptionV3, Xception, MobileNetV2, EfficientNetB0, and EfficientNetB1 were tested. Based on experimental results, the final RiceMultiNet model combining EfficientNetB1, ResNet50, and MobileNetV2 achieved a remarkable accuracy of 99.73%. As shown in **Table 5**, ensemble or hybrid approaches consistently outperform single models.

The application of preprocessing, augmentation, and normalization techniques reduced overfitting and improved generalization, enabling this high performance. These findings suggest that our approach is not only effective in a controlled setting but also highly suitable for practical use, helping farmers quickly and accurately identify diseases to protect crop health and enhance yields.

6.1. Limitations

There are still some significant drawbacks even though the suggested RiceMultiNet model demonstrated strong generalization and extremely high accuracy (99.37%). The dataset was first gathered from Kaggle. In actuality, elements like harsh lighting, intricate backgrounds, or partially harmed leaves may make disease detection more challenging. Second, it took a lot of time and effort to filter out low-quality photos and maintain class balance during dataset preparation. Third, even when using GPU resources, runtime problems like frequent resets or system hangs frequently interrupted Kaggle training. The entire experimentation process was slowed down as a result. Lastly, farmers or small research groups might not always have the relatively high computational power (GPU) needed for the model's training. The system's resilience and adaptability would be further enhanced in the future by gathering more varied and region-specific field photos.

6.2. Future Work

The robustness of the model will be enhanced in the future by gathering more varied rice leaf photos from various locales and actual field circumstances. Farmers and small research groups may find the system more useful if lightweight models appropriate for low-resource devices

are incorporated. Furthermore, creating a web-based platform or mobile application would make it easier for users to access the disease detection system in real time, increasing its usefulness for real-world field applications.

Data availability

We have made all resources publicly available to the academic community through the following Kaggle dataset URL: <https://www.kaggle.com/datasets/mirshaon/rice-leaf-diseases-zip>

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